

All-Degree Rigidity of a Weighted Keller Class

Classification, degenerations,
and an obstructed double point

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Abstract

We classify, without any bound on ordinary degree, a class of polynomial Keller maps in three variables naturally singled out by the Jacobian project in the `ProveIt` repository. The source weights are $(-1, 1, 2)$ and the target weights are $(2, 1, -1)$. In invariant coordinates $t = xy$ and $v = x^2z$, we assume that the first two numerators are affine in v and that the third numerator is affine in (t, v) . Every normalized map is either an explicitly invertible tame map or a member of a two-parameter diagonal orbit of the known Alpöge map. In the latter case its degree profile is forced to be $(7, 6, 4)$ and its expanded support has exactly sixteen monomials.

The proof replaces a degree-bounded coefficient search by one-variable polynomial rigidity. An equivariant source shear extends the classification to arbitrary polynomial dependence of r on t when its v -coefficient is constant, and gives the exact noninvertible degree spectrum $\{7, 8, 10, 12, \dots\}$. We also identify the reduced closure of the nonconstant branch as an affine plane whose coordinate axes are precisely its invertible members. Finally, for the normalized degree-seven coefficient problem with third numerator $1 + t + v$, we prove that the entire coefficient scheme is a double point, $\text{Spec } \mathbb{Q}[\varepsilon]/(\varepsilon^2)$. An explicit nonzero infinitesimal deformation is obstructed at second order. All finite identities and obstruction certificates are independently reproduced by an included exact-arithmetic script.

Status. Human-readable proofs and reproducible symbolic certificates; not a Lean/Rocq formalization or a peer-reviewed publication. The results extend the inspected repository statements, but publication priority is not established. This report does not claim a new counterexample to the Jacobian conjecture or a classification of unrestricted Keller maps.

[write] Batch 36, manuscript 01, placed in `1a1396d4d`: a report of ProveIt’s research-report collection that continues the formal project `Algebra/JacobianConjecture`. AI-assisted, unrefereed, not formalized; labels carry the prefix `wkr:`. The weighted determinant identity and the collision mechanism are the project’s own (Section 1.4); Gao’s three-dimensional map has *geometric* degree four (Section 1.5). Provenance: Appendix D. [Added 29 September 2026, batch 44:] Part II (Section 12, labels `wkr:as:`, not formalized) answers Section 10.2.

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1 Research question, provenance, and contribution

A useful research question should distinguish a structural theorem from a large but finite search. The Jacobian component of `ProveIt` provides a particularly concrete opportunity: it gives explicit maps, exact collision certificates, a weighted determinant reduction, and degree-bounded searches for simpler representatives. We ask whether one of its natural coefficient classes is rigid *without imposing a degree bound*. We then ask what remains of the apparent uniqueness when coefficients are allowed to contain nilpotents. These are separate questions: a unique field-valued solution need not be a reduced point of its coefficient scheme.

1.1 The repository baseline

The inspected snapshot is commit

e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9.

The relevant files are the Jacobian project README, its research README, and `verify_equivariant_sparsity.py`; immutable references are supplied in the bibliography [1, 2, 3].

The repository’s three-variable map can be written

$$\begin{aligned} u &= 1 + xy, & h &= u^2 z + y^2(1 + 3u), \\ F_0(x, y, z) &= (uh, y + 3xh, x(5 - 3u - x^2 z)). \end{aligned} \tag{1.1}$$

Its Jacobian determinant is -2 , and

$$F_0(-1, 1, 5) = F_0(0, -2, -16) = (0, -2, 0). \tag{1.2}$$

These identities, their formal verification, and their provenance are already part of the repository [1]. We verify the displayed identities again in the companion script but do not present them as new.

The research files restrict a sparsity calculation to maps of ordinary degree at most seven whose invariant numerators p, q are affine in v and whose third numerator r is affine in (t, v) . After normalizing $r = 1 + t + v$, the script proves that eleven optional coefficients cannot vanish, and identifies the branch $q_{40} = 0$ with the normalized map. Its certificate contains a square $(q_{21} - 4)^2$, but the repository discussion does not identify the entire coefficient scheme as the dual numbers [2, 3].

1.2 What this article establishes

The main result, Theorem 3.1, removes the ordinary degree restriction from that structural class. It also handles constant r and proves that the two one-term nonconstant cases are impossible. The proof forces the normalized nonconstant map outright, so it proves $q_{40} = 0$ rather than imposing it.

Theorem 6.3 extends the result beyond affine r by an explicit triangular normal form. Its degree-spectrum corollary determines all possible maximum degrees in that larger class. Theorem 7.1 describes the reduced two-parameter closure and its invertible boundary. Theorem 8.1 strengthens uniqueness in the fixed normalized degree-seven slice to an exact scheme presentation, and Theorem 8.4 identifies the full scheme on the nonconstant-parameter open locus. These results connect three levels of information:

Level	Conclusion proved here
Field-valued maps	An all-degree automorphism/counterexample dichotomy.
Reduced families	No deformation after fixing $r = 1 + t + v$; a parameter plane before fixing its two nonconstant coefficients.
Nonreduced families	One square-zero parameter in the degree-seven slice, with a nonzero second-order obstruction.

1.3 Relation to contemporary literature and limits

Shaska studies graded Keller maps, their quotient geometry, and the role of the sign pattern of weights [5]. Gao gives a tangent-sweep construction and further explicit examples, including a three-dimensional map of degree four [6]. Our subject is narrower: rigidity within a specified invariant-numerator class, not the construction of the known map or a minimum-degree theorem for all three-variable maps. In particular, our forced degree seven is compatible with examples outside this class of lower degree.

The searched sources and the inspected repository justify the comparison above, not a universal priority claim. The mathematical contribution of this report is its stated and proved classification and deformation results. Whether equivalent formulations already occur elsewhere requires further bibliographic review. Nothing in the proofs depends on accepting a new external claim about the general Jacobian problem: the particular map, its determinant, and its collisions are explicit.

1.4 What the project already contains

[write] This subsection and the next were added when the manuscript was written into ProveIt. Two ingredients that the manuscript presents in its own words are already recorded in the project’s research notes [2], `Algebra/JacobianConjecture/Research/README.md`, lines 188–224, at the pin and unchanged at the placement commit:

- the lift $F = (p(t, v)/x^2, q(t, v)/x, x r(t, v))$ with $t = xy$, $v = x^2z$, the polynomiality conditions of (2.3) in their general form, *the same determinant identity* (2.4) with the same chain-rule proof in the localization at x (Proposition 2.2), and the normalization (2.6) with determinant -1 (lines 188–208);
- the collision mechanism: with $\alpha = [t^2]p$ (the notes write a , which is a family parameter here), $F(0, y, z) = (\alpha y^2 + z, y, 0)$ and $F(1, t_0, v_0) = F(0, q(t_0, v_0), p(t_0, v_0) - \alpha q(t_0, v_0)^2)$ whenever $r(t_0, v_0) = 0$ (lines 210–224). The collision (6.6) is this mechanism at $t_0 = 0$, $v_0 = -1/\gamma$.

The manuscript cites the research notes for its baseline but not at these two points; they are credited here. What is new relative to the project is the all-degree classification and everything after it: the project’s sparsity certificate (`verify_equivariant_sparsity.py`, described in lines 240–248 of the notes) covers ordinary degree at most seven with $q_{40} = 0$ left as the “remaining branch”, and says it is “deliberately not advertised as a global sixteen-monomial theorem”. Corollary 6.1 proves $q_{40} = 0$ and the support profile in all degrees, inside the class only. For constant r the notes invoke Moh’s plane theorem through degree 100 (lines 232–238), in a class that allows v^2 terms; case (A) proves tameness in all degrees without Moh, but only for p, q affine in v , so the two statements sit side by side.

Formal status. The project’s Lean and Rocq/Coq developments formalize the map F_0 (Lean `counterexample`, `Algebra/JacobianConjecture/Lean/JacobianConjecture/C`

ounterexample.lean, lines 95–101), its determinant -2 (jacobianDet_counterexample), the collision (1.2) (collision), the weight- $(-1, 1, 2)$ torus action (counterexample_scaling, Algebra/JacobianConjecture/Lean/JacobianConjecture/Scaling.lean), the mirror symmetry and the refutation of the three-dimensional conjecture; the Coq development proves the corresponding statements. None of the weighted determinant identity, the normalization (3.4), the family $F_{a,b}$ or any theorem of this report is formalized.

1.5 Gao’s three-dimensional map has geometric degree four

[write] The “three-dimensional map of degree four” of [6] cited above and in Section 10 is Gao’s map of his Theorem 3.5, written G_{Gao} here because $G(t)$ below is the coefficient of v in r (after Theorem 6.3 and in Section 10), and its *degree four is the geometric degree*, the number of preimages of a generic point, the sense of “degree” defined in his abstract. Its three components have ordinary degrees 4, 11 and 12 and its Jacobian determinant is 2: this was read from his Section 3.5 at the write and checked there with SymPy 1.14.0 from his printed formulas (component degrees 4, 11, 12 and $\det J_{G_{\text{Gao}}} = 2$ as polynomial identities). So G_{Gao} is *not* a three-dimensional Keller map of lower ordinary degree than seven. The sentence “our forced degree seven is compatible with examples outside this class of lower degree” remains true as a compatibility statement, but G_{Gao} is not such an example; for Alpöge’s map Gao’s Theorem 3.3 gives geometric degree three. The project’s framing is unaffected: its notes say that the finite searches “do not prove that ordinary degree seven or sixteen monomials is globally minimal in dimension three” and that a simpler three-variable map would have degree at least four and, below seven, would break the weighted symmetry (Algebra/JacobianConjecture/Research/README.md, lines 12–15 and 262–269). Whether such a map exists is not settled by any source cited here.

2 Weighted coordinates and the exact determinant reduction

Throughout the classification, K is an arbitrary field of characteristic zero. All polynomial equalities are formal equalities over K .

Definition 2.1. A polynomial map $F : \mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ is a *Keller map* if its formal Jacobian determinant is a nonzero element of K . We use the source action

$$(x, y, z) \mapsto (\rho^{-1}x, \rho y, \rho^2 z)$$

and the target action with weights $(2, 1, -1)$. Equivariance means that the three coordinate polynomials are weighted homogeneous of weights 2, 1, -1 , respectively.

2.1 Invariant numerators

Set

$$t = xy, \quad v = x^2 z. \tag{2.1}$$

Both are invariant under the source action. Every equivariant polynomial map has a unique expression in the localization at x of the form

$$F = \mathcal{L}(p, q, r) := \left(\frac{p(t, v)}{x^2}, \frac{q(t, v)}{x}, xr(t, v) \right), \quad p, q, r \in K[t, v]. \tag{2.2}$$

For example, a monomial $x^i y^j z^k$ of weight 2 satisfies $i = j + 2k - 2$; multiplying it by x^2 gives $t^j v^k$. The corresponding formulas for weights 1 and -1 are $i = j + 2k - 1$ and $i = j + 2k + 1$. They prove existence and uniqueness of the numerators monomial by monomial.

For general numerators, polynomiality requires that every monomial $t^i v^j$ in p satisfy $i + 2j \geq 2$, and every such monomial in q satisfy $i + 2j \geq 1$. In the class studied here, $\deg_v p, \deg_v q \leq 1$, so these conditions reduce exactly to

$$p(0, 0) = p_t(0, 0) = q(0, 0) = 0. \quad (2.3)$$

There is no additional restriction on r for polynomiality. The apparent fractions in (2.2) are therefore polynomial expressions whenever (2.3) holds in our class.

Proposition 2.2 (Weighted determinant identity). *For a polynomial lift (2.2),*

$$\det J_F = \mathcal{J}(p, q, r) := \det \begin{pmatrix} -2p & p_t & p_v \\ -q & q_t & q_v \\ r & r_t & r_v \end{pmatrix}, \quad (2.4)$$

where the right side is evaluated at $(t, v) = (xy, x^2z)$.

Proof. Compute in coordinates (x, t, v) on $x \neq 0$. The coordinate change $(x, y, z) \mapsto (x, xy, x^2z)$ has determinant x^3 . The derivative matrix of $(x^{-2}p, x^{-1}q, xr)$ with respect to (x, t, v) has rows

$$(-2x^{-3}p, x^{-2}p_t, x^{-2}p_v), \quad (-x^{-2}q, x^{-1}q_t, x^{-1}q_v), \quad (r, xr_t, xr_v).$$

Factoring x^{-2}, x^{-1}, x from its rows and x^{-1} from its first column leaves the matrix in (2.4) and contributes x^{-3} . The chain rule cancels this with x^3 . Localization is injective on $K[x, y, z]$, so the identity holds as a polynomial identity, including on $x = 0$. $\square$

[write] This identity, with this proof, and the normalization below are already in the project's research notes (Section 1.4).

2.2 Normalization is exhaustive

Suppose that (2.3) holds and $\mathcal{J}$ is a nonzero constant. Evaluating at $(t, v) = (0, 0)$ gives

$$\mathcal{J}(0, 0) = -p_v(0, 0)q_t(0, 0)r(0, 0). \quad (2.5)$$

All three factors are therefore nonzero. Dividing the three target coordinates by these factors normalizes

$$p_v(0, 0) = q_t(0, 0) = r(0, 0) = 1. \quad (2.6)$$

For a Keller map, the resulting constant determinant is -1 . This operation preserves the structural restrictions on p, q, r , ordinary degrees, support sizes, and invertibility or noninvertibility. Thus a classification in this normalization classifies the unnormalized maps up to independent nonzero target scalings. We can write

$$r = 1 + \beta t + \gamma v. \quad (2.7)$$

The word *normalized* below always includes (2.3) and (2.6); it does not implicitly assume $\beta = \gamma = 1$.

3 The complete all-degree classification

Define the following polynomials over $\mathbb{Q}[a, b, t, v]$:

$$p_{a,b} = v - \frac{8a}{9}t^2 - 2abtv + \frac{28a^2b}{27}t^3 + \frac{4a^2b^2}{3}t^2v - \frac{8a^3b^2}{27}t^4 - \frac{8a^3b^3}{27}t^3v, \quad (3.1a)$$

$$q_{a,b} = t + 9bv - 8abt^2 - 12ab^2tv + 4a^2b^2t^3 + 4a^2b^3t^2v, \quad (3.1b)$$

$$r_{a,b} = 1 + abt + ab^2v. \quad (3.1c)$$

Let $F_{a,b} = \mathcal{L}(p_{a,b}, q_{a,b}, r_{a,b})$.

Theorem 3.1 (All-degree classification). *Let $p, q, r \in K[t, v]$ satisfy $\deg_v p, \deg_v q \leq 1$, let r be affine in (t, v) , and suppose that their lift is polynomial and normalized as in (2.6). Then $\mathcal{J}(p, q, r) = -1$ if and only if exactly one of the following descriptions applies.*

(A) $r = 1$ and, for unique $\lambda \in K$ and $B \in t^2K[t]$,

$$p = v + B(t), \quad q = t + \lambda(v + B(t)). \quad (3.2)$$

Its lift is a tame polynomial automorphism.

(B) r is nonconstant, both β and γ in (2.7) are nonzero, and

$$(p, q, r) = (p_{a,b}, q_{a,b}, r_{a,b}), \quad a = \frac{\beta^2}{\gamma}, \quad b = \frac{\gamma}{\beta}. \quad (3.3)$$

Its lift is noninjective. Its coordinate degrees are $(7, 6, 4)$, and its coordinate support sizes are $(7, 6, 3)$.

In particular, there are no normalized Keller maps in this class with exactly one of β, γ nonzero. No ordinary degree bound is assumed.

We prove the classification in Sections 4–5. The invertibility, collisions, and support assertions are proved in Section 6. The forward direction is the substantive rigidity statement; the reverse direction follows either from the formulas there or by substitution into (2.4).

3.1 The normalized representative

Put $F_* = F_{1,1}$. Its numerators are

$$p_* = v - \frac{8}{9}t^2 - 2tv + \frac{28}{27}t^3 + \frac{4}{3}t^2v - \frac{8}{27}t^4 - \frac{8}{27}t^3v, \\ q_* = t + 9v - 8t^2 - 12tv + 4t^3 + 4t^2v, \quad r_* = 1 + t + v.$$

It is exactly a normalization of (1.1):

$$F_* = \text{diag}\left(-\frac{1}{2}, -\frac{3}{2}, \frac{1}{2}\right) \circ F_0 \circ \text{diag}\left(1, -\frac{2}{3}, -2\right). \quad (3.4)$$

Thus the nonconstant part of the classification identifies a known representative rather than producing an unrelated counterexample.

4 One-variable rigidity lemmas

The decisive step is that constant weighted Wronskians control degree even when arbitrarily high-degree polynomials are allowed initially.

Lemma 4.1 (A degree bound from a constant weighted Wronskian). *Let $B, D \in K[t]$ satisfy*

$$DB' - 2BD' = \kappa, \quad \kappa \in K^\times. \quad (4.1)$$

Then $D \neq 0$, $\deg D \leq 1$, and $\deg B \leq 2$. More precisely, either D is a nonzero constant and B has degree one, or D has degree one and $B = \eta D^2 + \mu$ for some $\eta, \mu \in K$, with $-2\mu D' = \kappa$.

Proof. Neither B nor D can be zero. If D is constant, (4.1) gives $B' = \kappa/D$, so B has degree one.

Now put $n = \deg D \geq 1$ and $m = \deg B$. Unless $m = 2n$, the leading term on the left has degree $m + n - 1$ and nonzero coefficient proportional to $m - 2n$. This uses characteristic zero. If $m > 2n$, it contradicts the constant right-hand side. If $m = 2n$, subtract from B the unique scalar multiple of D^2 that cancels its leading coefficient. This changes neither side of (4.1). The resulting polynomial B_0 is nonzero and has degree $m_0 < 2n$. If initially $m < 2n$, just take $B_0 = B$.

The leading term calculation for B_0 now gives $m_0 + n - 1 = 0$. As $m_0 \geq 0$ and $n \geq 1$, we obtain $n = 1$ and $m_0 = 0$. Hence $B = \eta D^2 + \mu$, and substitution gives $-2\mu D' = \kappa$. $\square$

Lemma 4.2 (A cubic–square relation). *Suppose $A, C \in K[t]$ are nonzero and*

$$2CA' - 3AC' = 0, \quad A(0) = 1. \quad (4.2)$$

Then there are unique $E \in K[t]$ with $E(0) = 1$ and $c \in K^\times$ such that

$$A = E^3, \quad C = cE^2. \quad (4.3)$$

Proof. The derivative of A^2/C^3 is zero. A rational function over a characteristic-zero field with derivative zero lies in K : writing it in coprime numerator–denominator form, the identity $f'g - fg' = 0$ forces each nonconstant numerator or denominator to divide its own derivative, which is impossible by degree. Thus $A^2/C^3 \in K^\times$.

For every irreducible polynomial π , unique factorization gives $2 \operatorname{ord}_\pi A = 3 \operatorname{ord}_\pi C$. Consequently the multiplicities in A are multiples of three and those in C are the corresponding multiples of two. Because $A(0) = 1$, none of the common factors vanishes at zero. Normalize their product to have value one there. The resulting polynomial E satisfies $A = E^3$ and $C = C(0)E^2$. No extraction of a scalar root or extension of K is necessary. Uniqueness follows from the factorization and the value at zero. $\square$

Lemma 4.3 (The remaining common factor is linear). *Suppose that $A(0) = 1$, $B(0) = -1$, and*

$$DB' - 2BD' = -1, \quad (4.4)$$

$$DA' + 2CB' - 3AD' - 2BC' = 0, \quad (4.5)$$

$$2CA' - 3AC' = 0. \quad (4.6)$$

If $C \neq 0$, then in (4.3) the polynomial E has degree at most one.

Proof. Lemma 4.1 bounds $\deg D \leq 1$ and $\deg B \leq 2$. Lemma 4.2 gives $A = E^3$, $C = cE^2$. Suppose $m = \deg E \geq 2$.

If D has degree one, the leading coefficient of $DA' - 3AD'$ in degree $3m$ is $3(m-1)$ times the leading coefficient of D times the cube of that of E . It is nonzero. The other part $2CB' - 2BC'$ of (4.5) has degree at most $2m + 1 < 3m$, a contradiction.

If D is constant, it is nonzero and B has degree one. The term DA' has degree $3m - 1$, whereas $2CB' - 2BC'$ has degree at most $2m$. Again the leading term cannot cancel. Thus $\deg E \leq 1$. $\square$

Remark 4.4. The lemmas address the possible high-degree cancellations explicitly. A coefficient search stopped at any fixed degree would not establish these bounds. Characteristic zero is essential to the leading coefficient arguments and to the assertion about rational functions with zero derivative.

5 Proof of the classification, including exceptional cases

5.1 The case $\gamma \neq 0$: a moving coordinate

First reduce $r = 1 + \beta t + \gamma v$ to coefficient one on v . Replace

$$\tilde{p}(t, v) = \gamma p(t, v/\gamma), \quad \tilde{q}(t, v) = q(t, v/\gamma), \quad \tilde{r}(t, v) = r(t, v/\gamma). \quad (5.1)$$

The determinant and normalization are unchanged, while $\tilde{r} = 1 + \beta t + v$. We temporarily omit the tildes.

Set $s = 1 + \beta t + v$ and write uniquely

$$p = A(t)s + B(t), \quad q = C(t)s + D(t), \quad r = s. \quad (5.2)$$

Substitution into (2.4) gives

$$\mathcal{J} = E_0 + E_1 s + E_2 s^2, \quad (5.3)$$

where the expressions E_0, E_1, E_2 are the left sides of (4.4)–(4.6). For clarity, differentiation at fixed v gives $p_t = A's + A\beta + B'$, $q_t = C's + C\beta + D'$, and $r_t = \beta$. Expanding the determinant cancels the β terms and yields (5.3). Since (t, s) are independent polynomial coordinates, the Keller equation is exactly (4.4)–(4.6).

The boundary conditions become

$$\begin{aligned} A(0) &= 1, & B(0) &= -1, & A'(0) + \beta + B'(0) &= 0, \\ C(0) + D(0) &= 0, & C'(0) + \beta C(0) + D'(0) &= 1. \end{aligned} \quad (5.4)$$

We must not divide by C before excluding $C = 0$. If C were zero, the fourth condition would give $D(0) = 0$. Equation (4.4) at zero would then give $D'(0) = -1/2$, while (4.5) at zero would give $-3D'(0) = 0$. This is impossible.

The rigidity lemmas now apply. Write

$$\begin{aligned} E &= 1 + kt, & A &= E^3, & C &= cE^2, & c &\neq 0, \\ D &= dt - c, & B &= -1 + b_1 t + b_2 t^2. \end{aligned} \quad (5.5)$$

The first derivative condition in (5.4) gives

$$b_1 = -\beta - 3k. \quad (5.6)$$

After dividing (4.5) by the nonzero polynomial E , it reads

$$2c(EB' - 2kB) = 3(kc + d)E. \quad (5.7)$$

Its constant and linear coefficients, together with (5.6), give

$$b_2 = -\beta k - 2k^2, \quad \frac{d}{c} = -\frac{2\beta + 5k}{3}. \quad (5.8)$$

The final condition in (5.4) is $c(\beta + 2k) + d = 1$, which therefore simplifies to

$$c(\beta + k) = 3. \quad (5.9)$$

The coefficient of t in (4.4) is $-db_1 - 2cb_2$. Its vanishing, using (5.6) and (5.8), is

$$\frac{c}{3}(\beta + k)(2\beta + 3k) = 0.$$

Equation (5.9) permits cancellation of the first factors. Hence

$$k = -\frac{2\beta}{3}, \quad \beta \neq 0, \quad c = \frac{9}{\beta}, \quad d = 4, \quad b_1 = \beta, \quad b_2 = -\frac{2\beta^2}{9}. \quad (5.10)$$

The remaining constant coefficient of (4.4) is indeed -1 : $-cb_1 + 2d = -9 + 8 = -1$. All three equations hold for these values. We have proved uniqueness and existence, including the exclusion of $\beta = 0$.

More compactly, before undoing (5.1), the answer is

$$\begin{aligned} E &= 1 - \frac{2\beta}{3}t, & s &= 1 + \beta t + v, \\ p &= E^3 s - 1 + \beta t - \frac{2\beta^2}{9}t^2, & q &= \frac{9}{\beta}E^2 s + 4t - \frac{9}{\beta}. \end{aligned} \quad (5.11)$$

Replace v by γv and divide p by γ . Expansion yields (3.1) with $a = \beta^2/\gamma$ and $b = \gamma/\beta$. This proves the nonconstant branch whenever $\gamma \neq 0$.

5.2 The case $\gamma = 0$, $\beta \neq 0$: no solutions

Write $p = A(t)v + B(t)$, $q = C(t)v + D(t)$, and $r = 1 + \beta t$. The normalization implies $A(0) = 1$ and $B(0) = B'(0) = D(0) = 0$. The coefficient of v in (2.4) is

$$\beta AC + r(A'C - AC') = 0. \quad (5.12)$$

If $C \neq 0$, this says that $(rA/C)' = 0$, so $C = \lambda rA$ for some $\lambda \in K^\times$. When $C = 0$, the same expression holds with $\lambda = 0$. The constant coefficient of the determinant then becomes

$$A(\lambda(r^2 B)' - (rD)') = -1. \quad (5.13)$$

A product of two polynomials is a nonzero constant only when both are constants. Thus $A = 1$. Integrating (5.13) and using $B(0) = D(0) = 0$ gives

$$\lambda r^2 B - rD = -t. \quad (5.14)$$

Evaluate at $t = -1/\beta$. The left side is zero and the right side is $1/\beta \neq 0$. This contradiction excludes the entire case, at every degree.

5.3 The case $r = 1$: the full tame branch

Again write $p = Av + B$ and $q = Cv + D$. The coefficient of v in the determinant is $A'C - AC' = 0$. Since $A(0) = 1$, it follows that $C = \lambda A$ for some $\lambda \in K$, allowing $C = 0$. The constant coefficient is

$$A(\lambda B' - D') = -1.$$

Consequently $A = 1$ and $D = t + \lambda B$, using $B(0) = D(0) = 0$. Polynomiality and normalization give $B(0) = B'(0) = 0$, hence $B \in t^2K[t]$. This is precisely (3.2). Conversely, these formulas give $\mathcal{J} = -1$ directly. Together with the preceding two cases, this completes the algebraic classification in Theorem 3.1.

6 Invertibility, diagonal equivalence, and sharp support

6.1 An inverse for every constant-third-numerator map

Write $B(t) = t^2H(t)$. The lift of (3.2) is

$$F(x, y, z) = (P, y + \lambda xP, x), \quad P = z + y^2H(xy). \quad (6.1)$$

For output coordinates (U, V, W) , define

$$x = W, \quad y = V - \lambda WU, \quad z = U - (V - \lambda WU)^2H(W(V - \lambda WU)). \quad (6.2)$$

Substitution in both directions is immediate from $P = U$. This proves polynomial invertibility. More precisely, (6.1) is the composition of the elementary shear $z \mapsto z + y^2H(xy)$, the elementary shear $y \mapsto y + \lambda xz$ in the updated coordinates, and the coordinate permutation $(x, y, z) \mapsto (z, y, x)$. It is therefore tame. There is no bound on $\deg H$: arbitrarily large degrees occur on the invertible branch.

6.2 Every nonconstant map is in the same diagonal orbit

For $a, b \neq 0$, inspection of (3.1) gives

$$F_{a,b} = \text{diag}((ab^2)^{-1}, (ab)^{-1}, 1) \circ F_* \circ \text{diag}(1, ab, ab^2). \quad (6.3)$$

The product of the two diagonal determinants is one, consistently with $\det J_{F_{a,b}} = -1$. Thus there are two parameters before taking this source–target equivalence, and no parameter after taking it. The scalar parameters are uniquely recoverable from either r or two coefficients:

$$a = -\frac{9}{8}[t^2]p, \quad b = \frac{1}{9}[v]q. \quad (6.4)$$

Corollary 6.1 (Sharp all-degree support rigidity). *Within the class of Theorem 3.1, every noninvertible map has ordinary maximum degree seven and exactly sixteen expanded monomials. After ordering coordinates by their prescribed weights, its degree profile is $(7, 6, 4)$ and its support profile is $(7, 6, 3)$.*

Proof. Only branch (B) can be noninvertible. For $a, b \neq 0$, all coefficients displayed in (3.1) are nonzero. Different monomials in a numerator lift to different monomials. The degree of a lifted $t^i v^j$ is $2i + 3j - 2$ in the first coordinate, $2i + 3j - 1$ in the second, and $2i + 3j + 1$ in the third. The asserted profiles follow. Noninjectivity of each map is established just below. $\square$

This is stronger than saying that sixteen terms are minimal under a degree cap: every noninvertible map in the structural class has that precise support profile, regardless of the degree initially allowed. It is not a statement about minimum degree or support outside the class.

6.3 Collisions valid over the ground field

Let $\gamma = ab^2$ and $\ell = 9b$. Direct evaluation on the plane $x = 0$ gives

$$F_{a,b}(0, y, z) = \left(z - \frac{8a}{9}y^2, y, 0 \right). \quad (6.5)$$

At $t = 0$, the numerators simplify to $p = v$, $q = \ell v$, and $r = 1 + \gamma v$. Therefore

$$F_{a,b}\left(1, 0, -\frac{1}{\gamma}\right) = F_{a,b}\left(0, -\frac{\ell}{\gamma}, \frac{71}{\gamma}\right) = \left(-\frac{1}{\gamma}, -\frac{\ell}{\gamma}, 0\right). \quad (6.6)$$

To check the constant 71, note that $-(8a/9)\ell^2 = -72\gamma$; the first coordinate of the second point's image is therefore $(71 - 72)/\gamma$.

The two source points have different x coordinates. This proves noninjectivity over *every* characteristic-zero ground field K , without passing to an algebraic closure. It also gives a rational certificate when $a, b \in \mathbb{Q}^\times$.

Equivariance propagates this to every $\rho \in K^\times$:

$$\begin{aligned} F_{a,b}\left(\rho, 0, -\frac{1}{\gamma\rho^2}\right) &= F_{a,b}\left(0, -\frac{\ell}{\gamma\rho}, \frac{71}{\gamma\rho^2}\right) \\ &= \left(-\frac{1}{\gamma\rho^2}, -\frac{\ell}{\gamma\rho}, 0\right). \end{aligned} \quad (6.7)$$

These are explicit certificates in the coordinates of this report. They are consequences of the classified family, not a claim that the collision mechanism of the original map was unknown. [write] The mechanism is the project's (`Algebra/JacobianConjecture/Research/README.md`, lines 210–224): (6.6) is its case $t_0 = 0$, $v_0 = -1/\gamma$, and the second point $(0, q, p - \alpha q^2)$ with $\alpha = [t^2]p_{a,b} = -8a/9$ (the notes' a) is the one displayed. For $a = b = 1$ this was checked at the write by exact evaluation.

Example 6.2 (The normalized rational witness). For $a = b = 1$, the points $(1, 0, -1)$ and $(0, -9, 71)$ have common image $(-1, -9, 0)$. The Jacobian is nonsingular at both points because it has constant determinant -1 . Thus local nonsingularity and global injectivity are genuinely different conditions in the displayed family.

6.4 A larger class: arbitrary dependence on the first invariant

The affine restriction on r can be relaxed substantially without changing its essential normal form. The relevant source automorphisms are

$$T_h(x, y, z) = (x, y, z + y^2 h(xy)), \quad h \in K[t]. \quad (6.8)$$

They preserve the weights, have Jacobian one, and induce $(t, v) \mapsto (t, v + t^2 h(t))$. Their inverses are T_{-h} .

Theorem 6.3 (All-degree triangular extension). *Let p, q be affine in v and let*

$$r = R(t) + \gamma v, \quad R \in K[t], \quad \gamma \in K,$$

with polynomial lift and normalized origin coefficients. In particular, $R(0) = 1$. Then the normalized lift is Keller if and only if one of the following holds.

- (i) $\gamma = 0$, $R = 1$, and the map is a tame map (6.1).
- (ii) $\gamma \neq 0$, $\beta := R'(0) \neq 0$, and the map is

$$F = F_{a,b} \circ T_h, \quad a = \frac{\beta^2}{\gamma}, \quad b = \frac{\gamma}{\beta}, \quad h(t) = \frac{R(t) - 1 - \beta t}{\gamma t^2}. \quad (6.9)$$

Here h is a polynomial, uniquely determined by r . The map is noninjective over K .

Thus every noninvertible map in this larger class is the classified representative up to diagonal scalings and one explicit equivariant source shear. There is no degree bound on R , p , or q .

Proof. When $\gamma \neq 0$, the numerator defining h vanishes to order at least two at zero. Composing the source with T_{-h} replaces v by $v - t^2 h(t)$ and changes the third numerator to $1 + \beta t + \gamma v$. The first two numerators remain affine in v ; polynomiality and all normalized origin coefficients are preserved. The source shear has Jacobian one. Theorem 3.1 therefore applies: it forces $\beta \neq 0$ and identifies $F \circ T_{-h}$ with $F_{a,b}$. This is (6.9). The converse follows from the same composition, and collision points are transported by T_{-h} .

When $\gamma = 0$, use the proof of Section 5 with $r = R(t)$ instead of $1 + \beta t$. The coefficient of v is $r'AC + r(A'C - AC') = 0$, whence $C = \lambda rA$, including $C = 0$. The constant equation is still $A(\lambda(r^2B)' - (rD)') = -1$. It forces $A = 1$ and, on integration with the origin conditions, $\lambda r^2B - rD = -t$. Thus r divides t . Since $r(0) = 1$, this forces r to be constant and hence $r = 1$. The already proved tame classification finishes this case. $\square$

Corollary 6.4 (An exact degree spectrum). *In the noninvertible branch of Theorem 6.3, $h = 0$ gives coordinate degrees $(7, 6, 4)$. If $h \neq 0$ has degree m , the coordinate degrees are exactly*

$$(2m + 8, 2m + 7, 2m + 5). \quad (6.10)$$

Consequently the possible maximum ordinary degrees in this noninvertible class are precisely 7 and every even integer at least 8.

Proof. The coefficients of z in the three coordinates of $F_{a,b}$ have ordinary degrees 6, 5, 3, respectively. Their leading terms are nonzero for $a, b \neq 0$. The substitution $z \mapsto z + y^2 h(xy)$ adds terms of degrees $6 + (2 + 2m)$, $5 + (2 + 2m)$, and $3 + (2 + 2m)$. These are all strictly larger than the original coordinate degrees, so they cannot cancel against an original term. Their leading coefficients are nonzero, proving (6.10). Every $m \geq 0$ occurs by taking $h = t^m$. $\square$

The first invariant coefficient is the important distinction here. The coefficient of v in r is still required to be a scalar, not a nonconstant polynomial in t . No assertion is made for $r = R(t) + G(t)v$ with nonconstant G . [Added 29 September 2026, batch 44: when p and q are affine in v , a nonconstant G is impossible (Part II, Theorem 15.1), so Theorem 6.3 classifies every such map with r affine in v (Theorem 15.2).]

7 Reduced parameter geometry and invertible degenerations

The formulas (3.1) contain no negative powers of a or b . They therefore define a polynomial family on the entire parameter plane, not just on its torus. The diagonal formula (6.3) should not be used on the axes, where its individual scaling matrices are undefined. The polynomial formulas are the correct extension there.

Theorem 7.1 (The parameter plane and its boundary). *In the finite coefficient space of normalized weighted maps of ordinary degree at most seven with p, q affine in v and r affine in (t, v) , the following statements hold over K .*

- (i) The morphism $(a, b) \mapsto F_{a,b}$ is a closed embedding of $\mathbb{A}_K^2$. Every member has Jacobian determinant -1 .
- (ii) Its torus $ab \neq 0$ is exactly the normalized nonconstant- r branch. Its reduced Zariski closure is the entire embedded affine plane.
- (iii) The invertible members of this plane are exactly its two coordinate axes $V(ab)$. Explicitly,

$$F_{0,b}(x, y, z) = (z, y + 9bxz, x), \quad (7.1)$$

$$F_{a,0}(x, y, z) = \left(z - \frac{8a}{9}y^2, y, x \right). \quad (7.2)$$

- (iv) Inside this plane, the scheme-theoretic condition $r = 1$ cuts out the reduced nodal union $\text{Spec } K[a, b]/(ab)$.

Proof. The coefficients (6.4) recover a, b by linear functions on the ambient coefficient space. Every other coefficient is a polynomial in those two functions by (3.1). Thus the image is the graph of polynomial functions, a closed embedding.

The determinant is -1 on the torus by the classification and its explicit solution. It is a polynomial in a, b, t, v , so the identity holds on the whole plane. Equivalently, it can be checked directly from (3.1). The torus is dense, and the classification identifies all its field-valued nonconstant- r points. This proves the closure statement.

On the axes, the formulas reduce to (7.1) and (7.2), which are tame maps of the already classified constant branch. Off the axes, (6.6) excludes invertibility. Finally, imposing $r = 1$ on the parameter ring imposes $(ab, ab^2) = (ab)$. The ideal (ab) is radical: it is the intersection $(a) \cap (b)$ in $K[a, b]$. $\square$

7.1 What degenerates, and what does not

The Jacobian determinant stays -1 throughout this degeneration. The ordinary degrees and the collision coordinates do not stay bounded in the same manner. Fixing $b = 1$ and letting a approach zero over $\mathbb{R}$ or $\mathbb{C}$, the coefficients tend to the automorphism $F_{0,1}$, whereas the two source points in (6.6) have coordinates with poles in a . In this explicit sense the displayed collisions escape to infinity. No bounded limiting pair of distinct colliding points is asserted.

There are many other constant- r maps, because $B \in t^2 K[t]$ in (3.2) is arbitrary. Only the two axes described above lie in the closure of the nonconstant branch in this coefficient space. In particular, passing from the nonconstant branch to its closure does not recover the entire infinite-dimensional tame family.

7.2 Rigidity over every reduced coefficient algebra

Field-valued uniqueness can be strengthened without any degree cap on the numerators.

Corollary 7.2 (All-degree reduced rigidity). *Let R be a reduced $\mathbb{Q}$ -algebra. Suppose $p, q \in R[t, v]$ are affine in v , satisfy the normalization and polynomiality conditions, and satisfy $\mathcal{J}(p, q, 1 + t + v) = -1$. Then $p = p_*$ and $q = q_*$.*

Proof. For every prime ideal $\mathfrak{p}$ of R , pass to the fraction field of $R/\mathfrak{p}$. It is a characteristic-zero field, so Theorem 3.1 forces $a = b = 1$. Every coefficient of $p - p_*$ and $q - q_*$ therefore belongs to every prime ideal of R , hence to its nilradical. That nilradical is zero because R is reduced. $\square$

Thus fixing $r = 1 + t + v$ eliminates nonconstant families over reduced bases, even when large ordinary degrees are allowed. It does *not* eliminate families over nonreduced bases. We now determine one such coefficient scheme exactly.

8 The normalized coefficient scheme is a double point

The preceding proof used degree arguments over fields. Nilpotent leading coefficients need not satisfy those arguments. Scheme structure must therefore be analyzed rather than inferred from the unique field-valued solution.

8.1 The precise finite coefficient problem

Fix $r = 1 + t + v$ and ordinary degree at most seven. The most general normalized numerators in our class are

$$p = v + p_{20}t^2 + p_{11}tv + p_{30}t^3 + p_{21}t^2v + p_{40}t^4 + p_{31}t^3v, \quad (8.1)$$

$$q = t + q_{01}v + q_{20}t^2 + q_{11}tv + q_{30}t^3 + q_{21}t^2v + q_{40}t^4. \quad (8.2)$$

The degree formulas in the proof of Corollary 6.1 show that this list is exhaustive. In particular, the q_{40} term is allowed and must not be set to zero beforehand.

Let the coefficient vector, in the following fixed order, be

$$c = (p_{20}, p_{11}, p_{30}, p_{21}, p_{40}, p_{31}, q_{01}, q_{20}, q_{11}, q_{30}, q_{21}, q_{40}). \quad (8.3)$$

Let $I \subset \mathbb{Q}[c]$ be generated by all coefficients in t, v of $\mathcal{J}(p, q, 1 + t + v) + 1$. There are eighteen nonzero coefficient polynomials. Their complete expressions, ordering, and derivative matrix are supplied in `verification.json`; the article fixes enough data below to reconstruct and check the certificates independently.

Theorem 8.1 (Exact double-point presentation). *The full coordinate algebra of this coefficient problem is*

$$\mathbb{Q}[c]/I \simeq \mathbb{Q}[\varepsilon]/(\varepsilon^2). \quad (8.4)$$

Under this isomorphism $\varepsilon = q_{21} - 4$ and

$$c = c_* + u\varepsilon, \quad (8.5)$$

where

$$c_* = \left(-\frac{8}{9}, -2, \frac{28}{27}, \frac{4}{3}, -\frac{8}{27}, -\frac{8}{27}, 9, -8, -12, 4, 4, 0 \right) \quad (8.6)$$

and

$$u = \frac{1}{144}(-7, -18, 18, 24, -8, -8, 162, -198, -324, 144, 144, 0). \quad (8.7)$$

In particular, the entire ideal, not merely its radical, is generated by ε^2 and the eleven linear relations obtained from (8.5) by omitting the tautological q_{21} relation.

The proof has three components: uniqueness of geometric points, a rank certificate for the tangent space, and a certificate excluding a second-order lift. None requires a Gröbner-basis computation.

8.2 Tangent space and its exact rank certificate

Write the coefficient equations as $f_1, \dots, f_{18}$ in the order of the monomials

$$\begin{aligned} t^7, t^6v, t^6, t^5v, t^5, t^4v, t^4, t^3v^2, t^3v, t^3, \\ t^2v^2, t^2v, t^2, tv^2, tv, t, v^2, v. \end{aligned} \quad (8.8)$$

Let

$$M = \left(\frac{\partial f_i}{\partial c_j}(c_*) \right)_{i,j}. \quad (8.9)$$

Direct rational calculation gives $Mu = 0$. Delete the column corresponding to q_{21} and select the rows corresponding to

$$t^7, t^6, t^5, t^4v, t^4, t^3, t^2v, t^2, tv^2, t, v. \quad (8.10)$$

The determinant of that 11×11 submatrix is

$$-18\,874\,368 \neq 0. \quad (8.11)$$

Thus $\text{rank } M = 11$ and $\ker M = \mathbb{Q}u$. This proves that the scheme has a one-dimensional Zariski tangent space at its sole geometric point. It does not yet determine the length of the point.

Because $f_i(c_*) = 0$ and $Mu = 0$, the assignment $c = c_* + u\varepsilon$ solves the equations over $\mathbb{Q}[\varepsilon]/(\varepsilon^2)$. Since $u_{q_{21}} = 1$, the resulting map from the coefficient algebra onto the dual numbers is surjective. In particular, the tangent direction is not merely a redundant parametrization of a reduced point.

8.3 A nonzero obstruction at second order

Define the quadratic residual vector

$$H = ([\varepsilon^2]f_i(c_* + u\varepsilon))_{i=1}^{18}. \quad (8.12)$$

Consider a possible lift to $\mathbb{Q}[\varepsilon]/(\varepsilon^3)$ of the form $c = c_* + u\varepsilon + w\varepsilon^2$. Its second-order equation is

$$Mw + H = 0. \quad (8.13)$$

There is an explicit row functional

$$L = (-205, 0, -46, 0, -40, 24, -8, 0, 4, 0, 0, 0, 0, 0, 0, 0, 0, 0) \quad (8.14)$$

satisfying

$$LM = 0, \quad LH = -\frac{1}{72}. \quad (8.15)$$

Applying L to (8.13) produces a contradiction. For a nonzero scalar multiple ηu , the obstruction pairing is $-\eta^2/72$, which is again nonzero over every characteristic-zero field. There is therefore no second-order lift of any nonzero first-order tangent direction. The matrix, vector, and pairing in (8.9)–(8.15) are checked exactly by the included script.

8.4 From tangent obstruction to the entire scheme

We record the small algebraic fact that turns these certificates into a scheme-theoretic theorem.

Lemma 8.2 (An isolated point with one obstructed tangent). *Let k be algebraically closed and let A be a finitely generated k -algebra with exactly one geometric point and residue field k . Suppose its tangent space has dimension one and no nonzero tangent homomorphism $A \rightarrow k[\varepsilon]/(\varepsilon^2)$ extends to $A \rightarrow k[\varepsilon]/(\varepsilon^3)$. Then $A \simeq k[z]/(z^2)$.*

Proof. The Nullstellensatz implies that the radical of zero is the unique maximal ideal $\mathfrak{m}$ [7]. The algebra is Noetherian. A finitely generated ideal consisting of nilpotent elements is nilpotent: if its generators have nilpotence exponents n_i , every product of more than $\sum_i (n_i - 1)$ generators is zero. Thus some power of $\mathfrak{m}$ is zero, and A is a finite-dimensional local k -algebra.

The cotangent space $\mathfrak{m}/\mathfrak{m}^2$ has dimension one. Choose $z \in \mathfrak{m}$ whose residue spans it. Nakayama's lemma gives $\mathfrak{m} = (z)$ [8]. Repeatedly writing an element as a scalar plus z times another element, and using nilpotence, shows that A is generated by z over k . Since $k[Z]$ is a principal ideal domain and the only root of the defining ideal is zero, it follows that $A \simeq k[Z]/(Z^N)$ for some N . The nonzero cotangent space forces $N \geq 2$. If $N \geq 3$, sending Z to ε defines a homomorphism to $k[\varepsilon]/(\varepsilon^3)$ with nonzero first-order part, contrary to the hypothesis. Hence $N = 2$. $\square$

Proof of Theorem 8.1. Work first over $\overline{\mathbb{Q}}$. The classification with $\beta = \gamma = 1$ forces $a = b = 1$, so the coefficient scheme has exactly one geometric point, c_* . Equations (8.11) and $Mu = 0$ give tangent dimension one, and (8.15) excludes every nonzero second-order lift. Lemma 8.2 gives a two-dimensional coordinate algebra over $\overline{\mathbb{Q}}$.

Already over $\mathbb{Q}$, substitution $c = c_* + u\varepsilon$ defines a surjection

$$\mathbb{Q}[c]/I \longrightarrow \mathbb{Q}[\varepsilon]/(\varepsilon^2).$$

After extension to $\overline{\mathbb{Q}}$, this is a surjection between vector spaces of dimension two and is an isomorphism. Field extension is faithful, so its kernel over $\mathbb{Q}$ was zero. This proves (8.4) with the specified coordinates.

The kernel of the explicit substitution map from the polynomial ring $\mathbb{Q}[c]$ is the ideal generated by ε^2 and the eleven affine-linear relations $c_j - (c_*)_j - u_j\varepsilon$. The isomorphism identifies this kernel with I , proving the exact ideal assertion. $\square$

8.5 The universal meaning of the answer

Corollary 8.3 (Solutions over arbitrary coefficient algebras). *For every $\mathbb{Q}$ -algebra R , the solutions of (8.1)–(8.2) with $\mathcal{J} = -1$ are in bijection with*

$$\{e \in R : e^2 = 0\}.$$

The corresponding solution is $c = c_ + ue$, and $e = q_{21} - 4$.*

Proof. Apply $\mathrm{Hom}_{\mathbb{Q}\text{-alg}}(-, R)$ to (8.4). $\square$

In particular, every solution over a reduced ring is the constant one, but a coefficient ring containing a nonzero square-zero element supports a nontrivial infinitesimal deformation. The parameter is obstructed rather than the initial segment of a moving curve of normalized field-valued solutions. The square seen in the repository's finite coefficient certificate is therefore genuine scheme structure, not just an artifact of one Gröbner-basis presentation.

8.6 The variable-parameter open scheme

The nonconstant coefficient parameters can also be restored without losing control of nilpotents. Let $\mathcal{X}^\times$ denote the coefficient scheme obtained by using (8.1)–(8.2), putting $r = 1 + \beta t + \gamma v$, and inverting $\beta\gamma$.

Theorem 8.4 (Torus times the double point). *There is an explicit isomorphism*

$$\mathcal{X}^\times \simeq \operatorname{Spec} \mathbb{Q}[\beta^{\pm 1}, \gamma^{\pm 1}, \varepsilon]/(\varepsilon^2). \quad (8.16)$$

It is an isomorphism of full schemes, not just of reduced varieties. The square-zero coordinate can be taken to be $\varepsilon = q_{21}/(\beta\gamma) - 4$.

Proof. Over any $\mathbb{Q}$ -algebra where β, γ are units, define

$$\bar{p}(t, v) = \gamma p(t/\beta, v/\gamma), \quad \bar{q}(t, v) = \beta q(t/\beta, v/\gamma). \quad (8.17)$$

The third numerator becomes $1 + t + v$. The two normalized origin coefficients remain one. These substitutions preserve the allowed monomial supports and ordinary degree bound, and have polynomial inverse formulas over the Laurent coefficient ring. They preserve the determinant condition: the row scalings contribute $\beta\gamma$ and the derivative column scalings contribute $(\beta\gamma)^{-1}$.

Thus (8.17) identifies the coefficient functor with a choice of two units and a solution of the fixed normalized scheme. Apply Theorem 8.1 to that fixed scheme. The coefficient of t^2v in $\bar{q}$ is $q_{21}/(\beta\gamma)$, giving the stated square-zero coordinate. All formulas are valid with nilpotent coefficients, so this proves the scheme isomorphism. $\square$

At every geometric point of this open scheme the reduced space has dimension two and the tangent space has dimension three: two tangent directions come from the torus and one from its square-zero thickening. The remaining boundary problem is how this structure meets the axes, where the Laurent normalization is no longer available.

Remark 8.5 (The degree boundary matters). Corollary 7.2 is an all-degree statement. In contrast, Theorem 8.1 fixes the ordinary degree-seven coefficient space. We have not proved that allowing higher-degree coefficients over nonreduced rings introduces no extra tangent directions or higher-order nilpotents. The field-valued classification alone does not imply that stronger assertion.

9 Decision procedure, reproducibility, and formalization

9.1 A classification-based recognition procedure

The theorem gives more than a determinant identity. For rational input polynomials in the structural class, it yields a direct recognition procedure for the Keller condition, together with an inverse or collision certificate.

First check the prescribed weights monomial by monomial and extract the unique invariant numerators as in Section 2. Check $\deg_v p, \deg_v q \leq 1$ and that r is affine. Inputs outside these restrictions are *outside the procedure's scope*; they are not rejected as non-Keller maps.

Within this scope, compute $p_v(0), q_t(0), r(0)$. If one is zero, the map is not Keller by (2.5). Otherwise normalize the target coordinates. Write $r = 1 + \beta t + \gamma v$. If $\beta = \gamma = 0$, take $B = p - v$ and $\lambda = [v]q$, and test the exact identities (3.2) with $B \in t^2K[t]$. If exactly one of β, γ is nonzero, the map is not Keller. If both are nonzero, recover a, b from (3.3) and compare

with the fixed formula (3.1). The classification proves both soundness and completeness of these comparisons inside the stated scope.

For a positive result, return either the inverse (6.2) or the pair of source points in (6.6). To undo target normalization, compose an inverse with the inverse target scaling; collision source points remain collision source points, with their common target rescaled. This avoids a search over all possible inverses or collision pairs.

The larger class of Theorem 6.3 is recognized by first computing its unique source shear. For $\gamma \neq 0$, compute h from (6.9), compose the source with T_{-h} , and apply the affine- r procedure. For $\gamma = 0$, a nonconstant $R(t)$ is excluded by that theorem. Inputs with nonconstant v -coefficient in r remain outside the classified scope. [Added 29 September 2026, batch 44: they are now classified, and rejected, by Part II's Theorem 15.1; the extended procedure is in Section 21.]

For canonical sparse input, reading and comparing the coefficient lists requires a number of field operations linear in the input support size, apart from the cost of canonicalizing the input. This is an arithmetic operation statement, not a claim of linear bit complexity for arbitrarily large rational coefficients. An implementation should retain that distinction.

9.2 What the companion script checks

The file `verify_results.py` ([write] shipped as `code/verify_results.py`, with its record in `data/`) works in exact rational and symbolic arithmetic. It reconstructs the equations rather than trusting a stored matrix. Its thirteen groups of checks cover:

1. The two-parameter determinant in both invariant coordinates and the full three-variable Jacobian; the original-map normalization and collision; diagonal equivalence, degrees, support, and the universal collision family.
2. The two degenerate axes, the two-sided inverse identity with a formal arbitrary function H , the source-shear identities and sample degree profiles, and the one-variable coefficient identities and explicit endpoint of the classification proof.
3. The invertible variable-parameter scheme normalization, the eighteen coefficient equations, the rank minor, the tangent vector, the cokernel functional, the second-order obstruction, and the vanishing of every coefficient equation under the dual-number presentation.

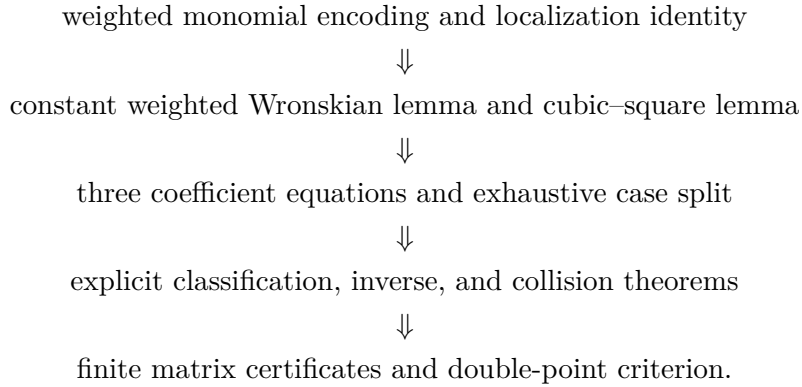
The last check proves one ideal containment by direct substitution. The reverse containment is justified in the article by the classification, rank, and obstruction argument; the script does not silently label a one-sided ideal check as equality.

The run included with this report used SymPy 1.14.0 and passed all thirteen check groups. The JSON certificate includes the full rational matrix and both row and column orderings. No numerical tolerance, random test, or floating-point approximation is used. These checks are reproducibility evidence for the finite algebra; they do not replace the universal one-variable degree proofs.

9.3 A realistic proof-assistant boundary

The new theorems in this report have not been compiled in Lean or Rocq. The existing repository formalizations of the original map are separate from the new classification. A

sensible formalization order is the following dependency chain:



The main classification needs polynomial degrees, characteristic-zero derivatives, a rational-function constant lemma, and unique factorization in one variable. It does not require the heavy geometric machinery used in the literature to study unrestricted Keller maps. The double-point statement additionally needs finite-type algebra, the Nullstellensatz, Nakayama's lemma, and faithful scalar extension, or an alternative direct ideal-membership certificate.

One attractive alternative for formalizing the last step is to produce explicit polynomial combinations expressing the twelve simple generators of the double-point ideal in terms of the eighteen original equations. The present argument proves these combinations exist but does not supply their expanded coefficients. Such certificates could isolate the formal scheme result from a large commutative-algebra library.

10 Further research questions and proposed next steps

The following questions are proposed by the present results. Unless explicitly attributed, they are research directions formulated here, not claims that a published source listed them as open. They are ordered to separate immediately testable extensions from much broader classification problems.

10.1 Allow quadratic dependence on the second invariant

Keep r affine but allow $\deg_v p, \deg_v q \leq 2$. Does the nonconstant branch admit genuinely new diagonal-equivalence classes? After the change $s = r$, the determinant becomes a polynomial of higher degree in s . The highest coefficient should impose a weighted differential relation on the leading coefficients, but the lower equations can couple several common factors. The first concrete task is to derive the complete coefficient system and identify whether an analogue of Lemma 4.1 gives an a priori degree bound. A finite search would be useful evidence, but an all-degree conclusion needs such a bound or another structural argument. [Added 29 September 2026, batch 44: still open. Part II's Proposition 20.4 shows that a v^2 term is needed, and suffices, to lift its obstructed slope directions at the tame point to second order; that is not a classification of this class.]

10.2 Allow a nonconstant second-invariant coefficient in r

Theorem 6.3 settles arbitrary polynomial t -dependence when the coefficient of v in r is constant. The next case is $r = R(t) + G(t)v$ with nonconstant $G(t)$. Must G be constant under suitable Keller and polynomiality hypotheses, or do genuinely new normal forms appear?

The substitution $s = r$ now requires division by G and is not a global polynomial coordinate change. Track the behavior at the zeros of G and determine whether the constant-Wronskian argument has a localized analogue. Allowing a quadratic term in v is a further, distinct extension. [Added 29 September 2026, batch 44: *answered*. With p, q affine in v , G must be constant, in all degrees (Theorem 15.1); the zeros of G are controlled by a rational pole equation (Lemma 16.2). No new normal forms appear (Theorem 15.2). The quadratic extension remains open (Section 10.1).]

10.3 Does the nonreduced structure stabilize with degree?

For each $d \geq 7$, form the normalized coefficient scheme with $r = 1 + t + v$, p, q affine in v , and lifted ordinary degree at most d . Its reduced scheme is a point by Corollary 7.2. Is the full scheme always the same double point? Or can high-degree coefficients support new nilpotent directions invisible at degree seven? A first step is to linearize the three identities in (A, B, C, D) around the explicit solution without truncating their degrees, solve the resulting polynomial linear differential system, and then examine its obstruction maps. This is a particularly well-targeted extension of the present work. [Added 29 September 2026, batch 44: still open. Part II's nilpotent slopes (Theorem 20.2, Corollary 20.6) live at the tame point with r free, not in this scheme with $r = 1 + t + v$ fixed; they neither extend nor contradict Theorem 8.1.]

10.4 Scheme-theoretic gluing at the invertible boundary

Theorem 8.4 settles the full degree-seven coefficient scheme on $\beta\gamma \neq 0$: it is the parameter torus times a double point. Determine how this thickening meets the constant- r tame locus where that normalization ceases to be invertible. Does its scheme-theoretic closure acquire embedded components or higher nilpotent order on either axis or at their intersection? The reduced affine-plane closure and the nodal boundary in Theorem 7.1 do not decide this gluing problem. A concrete first step is to saturate the coefficient ideal by $\beta\gamma$, then study its specialization on each axis with explicit ideal-membership certificates. [Added 29 September 2026, batch 44: still open; Part II's Theorem 20.3 constrains the tame end of the gluing (Section 22.5).]

10.5 Other weights and other Wronskian exponents

Our determinant produces a cubic-square relation because the first two weights are 2 and 1. For other mixed-sign weight triples, analogous highest-coefficient equations may have the form $mCA' - nAC' = 0$. Investigate the resulting common-power factorizations and constant weighted Wronskians. The interesting issue is not just repeating the calculation with new constants: the polynomiality inequalities at the coordinate hyperplane may either permit or rule out the required factorizations. A systematic analysis could delineate entire rigid classes of equivariant maps.

10.6 Compare lower-degree maps outside this symmetry class

The current literature already includes a three-dimensional degree-four example [6]; the existence of such an example is therefore not posed as a new open problem here. ([write] Gao's degree is the geometric degree; the ordinary degrees of his map are 4, 11, 12, see Section 1.5. A three-variable Keller map of *ordinary* degree below seven is not known from the sources cited here.) Instead, extract its invariant or geometric structure and determine precisely which hypothesis of our classification fails. Compare support size, rational coefficient height, and the

cost of a formal collision certificate. This would help replace a single “simplest map” question by explicit, noninterchangeable complexity criteria.

10.7 Arithmetic optimization within the diagonal orbit

Over $\mathbb{Q}$, the classification makes every nonconstant map in this class explicitly diagonal-equivalent. One can now optimize primitive coefficient height, denominator size, or integral collision height within that orbit. The parameters a, b and the additional unnormalizing target scalings make these concrete arithmetic optimization problems. Prove lower bounds for a specified normalization rather than relying on finite samples of small rational parameters.

10.8 Direct ideal certificates and kernel verification

Generate explicit membership certificates for the twelve simple generators in Appendix A, and check them with polynomial normalization in Lean or Rocq. This would turn the deformation theorem into a compact finite certificate after the all-degree classification has been formalized. The target should be a transparent theorem boundary: no hidden use of a computer-algebra claim that a Gröbner basis is complete.

10.9 Positive characteristic as a controlled contrast

The displayed family and finite identities have meaningful reductions at many primes, but the all-degree proof does not automatically survive. In positive characteristic, nonconstant p th powers have derivative zero, and the weighted degree coefficients can vanish. Classify which parts of the theorem persist under explicit degree bounds and which fail through Frobenius phenomena. Separately determine the primes at which the tangent rank or obstruction changes. Reducing a characteristic-zero certificate is not sufficient to claim the same unrestricted classification.

11 Conclusion

The repository’s finite sparsity problem admits a stronger structural answer in its affine-invariant class. Constant third numerator gives all and only the explicit tame maps (3.2). Nonconstant third numerator forces both invariant coefficients to be nonzero and forces the entire map into the diagonal orbit (3.1), with degree seven and sixteen terms. An explicit source shear extends the classification to $r = R(t) + \gamma v$ and gives the exact degree spectrum $\{7, 8, 10, 12, \dots\}$. The proof is an all-degree argument, not an extrapolation from a coefficient search.

The parameter-space results explain two different notions of rigidity. Before fixing the third numerator, the nonconstant branch has an affine plane closure with invertible coordinate axes. After fixing $r = 1 + t + v$, all reduced families are constant, yet the degree-seven coefficient scheme retains exactly one square-zero parameter. Its tangent direction has an explicit nonzero second-order obstruction. Restoring the two nonconstant parameters gives the torus times this double point. These distinctions are essential both for further searches and for a trustworthy formalization.

Part II. All-degree classification of affine-in- z weighted Keller maps

Sections 1–11 and Appendices A–D form the original report (Part I) of 24 September 2026, unchanged except for dated pointers to this addition. Part II was added on 29 September 2026 in batch 44 of ProveIt’s incoming-report intake, from one manuscript written against Part I as printed above. It answers the question of Section 10.2 affirmatively: when all three invariant numerators are affine in v , the Keller condition forces the coefficient of v in r to be constant, with no bound on ordinary degree. Section 12 records the provenance, fixes the notation against Part I and states what is answered and what is not.

12 Provenance, conventions and what Part II answers

12.1 One manuscript, one addition

Part II is batch-44 manuscript 04, dated 29 September 2026: *All-Degree Classification of Affine-in- z Weighted Keller Maps: Constant-slope rigidity and obstructed nilpotent deformations*. Its author line is “Prepared for Vladimir Reshetnikov” (PDF metadata: “AI-assisted research draft prepared for Vladimir Reshetnikov”). It was delivered as the archive `ProveIt_Affine_Weighted_Keller_Classification` (inner directory of the same name, main file `article.tex`, a 22-page A4 PDF). It is pinned to the ProveIt commit

9b24a3a8d545af9624f6ac455f5b548be62818b6

(committed 2026-09-29 16:02:05 UTC). The archive arrived in commit `ae28ea2db` and was placed in 203016015 as an *addition* to this report rather than as a new report, because it answers Part I’s Section 10.2 and restates Part I’s normal forms. Neither Part I nor the Jacobian project `Algebra/JacobianConjecture` changed between the pin and the placement commit, so the manuscript read exactly the text printed above.

Its companion files are shipped with the prefix `02-affine-slope-` (Section 25); its manuscript, README and PDF are not shipped, this Part being their written form. Every result, proof, example, table, question and limitation of the manuscript is printed; the few items that duplicate Part I are printed once (Section 12.4). Text added at the write, beyond the present section, is marked [write]. The manuscript’s title page and abstract read as follows.

Main result. In characteristic zero, the Keller condition forces the coefficient of $v = x^2z$ in the third invariant numerator to be constant, provided all three invariant numerators are affine in v . This closes the explicit extension question in Section 10.2 of ProveIt’s *All-Degree Rigidity of a Weighted Keller Class* (Part I). The resulting classification has no ordinary degree bound.

A second result. The field-level rigidity does not hold over dual numbers. Nonconstant nilpotent slopes exist in arbitrarily high degrees, but every nonzero slope direction at the simplest tame map is obstructed at second order, even when arbitrary affine-in- v second-order corrections are allowed.

Status. Complete written proofs and reproducible exact symbolic checks are supplied. The new results are not formalized in Lean or Rocq, not independently refereed, and not asserted to have established worldwide priority. The known counterexample and the previously classified constant-slope family are credited separately.

Abstract. Let K be a field of characteristic zero, let $t = xy$ and $v = x^2z$, and consider polynomial lifts

$$F(x, y, z) = (x^{-2}p(t, v), x^{-1}q(t, v), xr(t, v)), \quad \deg_v p, \deg_v q, \deg_v r \leq 1.$$

These are exactly the maps affine in z with source weights $(-1, 1, 2)$ and target weights $(2, 1, -1)$. We prove that a nonzero constant Jacobian forces r_v to be constant. The proof is not a bounded coefficient search: a highest-coefficient identity gives a cube–square factorization; a rational moving coordinate produces an equation $LL' = \alpha/U^3$; pole orders and polynomiality at the coordinate hyperplane then exclude every nonconstant slope. Combined with an explicit reconstruction, this gives all maps in the class: a tame family, or one fixed noninjective map up to diagonal scalings and a single equivariant source shear. We supply an exact coefficient-only recognition algorithm, inverses or collision certificates, and the degree spectrum $\{7\} \cup \{8, 10, 12, \dots\}$ for the noninjective branch.

We also determine a sharp limitation of this rigidity. It persists over reduced $\mathbb{Q}$ -algebras, but fails over dual numbers. At the map (z, y, x) , any nonzero first-order variation of the third slope has a nonvanishing second-order obstruction, independent of all second-order corrections inside the affine-in- z class. In bounded coefficient schemes, this gives nonzero nilpotent slope coordinates and arbitrarily large spaces of obstructed tangent directions. Full proofs, source comparisons, twelve exact-check groups, and a further research program are included.

12.2 What Part II answers, and what it does not

- *Section 10.2 is answered, affirmatively.* With p, q affine in v and $r = R(t) + G(t)v$ in Part I's notation, Theorem 15.1 proves that G is constant, with no bound on ordinary degree. Part I's Theorem 6.3 then applies, so Part I's classification covers every affine-in- z weighted Keller map (Theorem 15.2). The exclusions stated at the end of Section 6.4 and in Section 9 are lifted accordingly (dated pointers there).
- *Section 10.1 (v^2 terms in p, q) is not answered.* Proposition 20.4 shows that a v^2 term is needed, and suffices, to lift the canonical obstructed directions at the tame point to second order; it is not a classification of the quadratic class.
- *Section 10.3 is not answered.* The nilpotent results of Section 20 concern slope directions at the tame point $(v, t, 1)$, in coefficient schemes in which r varies; Theorem 8.1 concerns the noninjective point with $r = 1 + t + v$ fixed. They neither extend nor contradict the double point, as the manuscript says itself (Section 22.4).
- *Section 10.4 is not answered;* Theorem 20.3 is a constraint on it (Section 22.5).

12.3 Notation: Part I against Part II

Part II keeps the manuscript's letters, which were chosen before it was joined to Part I, so several letters mean different things in the two Parts. Table 1 fixes them; within Part II only the Part II column applies. The tempting false readings are printed in the last column.

Table 1: Symbols whose meaning differs between the Parts.

Symbol	Part I	Part II	Not to be read as
a, b	family parameters of $F_{a,b}$, $a = \beta^2/\gamma$, $b = \gamma/\beta$	coefficient polynomials, $p = a(t)v + b(t)$; Part I's parameters are written a_I, b_I	the parameters of (3.1)

Symbol	Part I	Part II	Not to be read as
c, d	scalars in $C = cE^2$, $D = dt - c$ (5.5); c also the coefficient vector, c_* its point	coefficient polynomials, $q = c(t)v + d(t)$	scalars
e, g	e : the square-zero element of Corollary 8.3	$r = g(t)v + e(t)$: $e = r(t, 0)$, and $g = r_v$ is the <i>third slope</i> ; they are Part I's $R(t)$ and $G(t)$ of Section 10.2	nilpotents
G	coefficient of v in r (Sections 6.4, 10.2); G_{Gao} is Gao's map	a first-order slope variation (Theorems 20.2, 20.3); the manuscript's valuation $G = \text{ord}(g)$ is written o_g ; Section 13 quotes Part I's G , which is g here	Gao's map
A, B, C, D	$p = As + B$, $q = Cs + D$ (5.2)	the same in (16.6)–(16.7), with rational $A, \dots, D$; first-order variations of a, b, c, d in Theorem 20.3. The manuscript's $A = \text{ord}(a)$, $C = \text{ord}(c)$, its collision points A, B , its reduced algebra A and its coefficient algebras A_D are written $o_a, o_c, X_1, X_2, \Lambda$ and Λ_D	the same objects in every section
U	an output coordinate (6.2)	a polynomial in t ; at the end $U = 1 - 2\beta t/3$, which is Part I's E of (5.11)	a coordinate
k	$E = 1 + kt$	the scalar in $c = kHU^2$; at the end $k = 9/\beta$, which is Part I's scalar c of (5.10)	Part I's $k = -2\beta/3$
H	$B = t^2 H(t)$ (6.1); the residual vector (8.12)	the cube root in $g = H^3$	either
s, S	$s = 1 + \beta t + v$	$s = e + H^3 v$ and $S = e + \gamma v$; each is r	a constant
$L, \mathcal{L}$	L : the row functional (8.14); $\mathcal{L}$: the lift	$L = \delta - \alpha T$, a rational function; $\mathcal{L}$ is the same lift	a functional
α	(write notes) the project's coefficient $[t^2]p$	$\alpha = 2k/3$ in (16.8)	$[t^2]p$
T	only in T_h	$T = B/w^2$ (Section 16); the indeterminate of the cubic (19.1); T_h as in Part I	one object
P, Q, R	$R(t)$: the t -part of r in Theorem 6.3	the components of F , and target coordinates in the cubic model; Section 13 quotes Part I's $R(t)$	Part I's $R(t)$, which is e here
D	—	the degree cap of Λ_D and the input degree of the algorithm (the manuscript writes D)	the polynomial D
τ	—	$y + 1/x$ (Section 19)	$t = xy$

The field, the weights, the invariants $t = xy$, $v = x^2 z$, the lift $\mathcal{L}$, the Jacobian $\mathcal{J}$, the normalization to determinant -1 , the pair β, γ (for $e = 1 + \beta t$ and $g = \gamma$), the source shear T_h and its polynomial h , the map F_0 and its normalization F_* , and the dual-number variable ε mean the same in both Parts. “Degree” means ordinary polynomial degree unless qualified as

function-field or geometric degree. Renamed from the manuscript: its field $\mathbb{K}$ is printed K ; its valuations A, C, G are o_a, o_c, o_g ; its collision points A, B are X_1, X_2 ; its reduced algebra A is Λ ; its degree cap D and coefficient algebras A_D are D and Λ_D ; Part I's parameters in its comparison sentence are a_I, b_I . No normalization was changed. Equations are numbered by section, as in Part I; the manuscript numbered them consecutively, (1)–(59), and its section k is Section $k + 12$ here.

12.4 What is printed once

The manuscript restates several items of Part I to be self-contained. They are printed once, and where they occur Part II points to them.

- The map F_0 , its determinant -2 and its integral collision (manuscript equations (1)–(2)) are (1.1), (1.2); the normalization $F_* = \text{diag} \circ F_0 \circ \text{diag}$ (manuscript equation (39)) is (3.4).
- The lift and the weighted Jacobian identity with its chain-rule proof (manuscript equation (4) and Proposition 2.2 with equation (7)) are (2.2) and Proposition 2.2, which are in turn the project's (Section 1.4). The manuscript's coefficient expansion (14.7)–(14.9) is new and printed.
- The tame inverse and its decomposition into two elementary shears and a permutation (manuscript Section 6.1, equations (34)–(35), with output coordinates (X, Y, Z)) are (6.1)–(6.2) (output coordinates (U, V, W)).
- The proof of the degree spectrum (manuscript Corollary 6.4) repeats Part I's proof of Corollary 6.4; the statement is printed as Corollary 18.4 because its scope is larger, and its proof refers to Part I.
- Printed although partly known, because they carry the argument: the determinant expansion (16.7), which is Part I's (5.3) with (4.4)–(4.6) (here with rational coefficients); Lemma 16.1, which Part I proves inside Lemma 4.2; the normal forms of Theorem 15.2, which restate Theorems 3.1 and 6.3 in the coordinates β, γ, e over the enlarged class (Remark 15.3); and the cubic model of Section 19, which the manuscript does not advertise as new (see the [write] note there).

The manuscript's reconstruction of the noninjective branch (Section 17) is a genuinely different route from Part I's Section 5: it reuses the moving-coordinate equations with $H = 1$ instead of the constant weighted Wronskian lemmas. It is printed in full as a second proof.

12.5 The formal project and Part II

Check group 4 of the companion script (`original_map`) rechecks in SymPy statements that the project has *kernel-checked* (Lean namespace `LeanProofs.JacobianCounterexample`; verified at the write in the files at the placement commit):

- $\det DF_0 = -2$: Lean `jacobianDet_counterexample (Algebra/JacobianConjecture/Lean/JacobianConjecture/Counterexample.lean, line 134)`, a polynomial identity over every commutative ring; Rocq `jacobian_det_is_minus_two (Algebra/JacobianConjecture/Coq/Counterexample.v, line 182)`, at every point;
- the collision $F_0(-1, 1, 5) = F_0(0, -2, -16) = (0, -2, 0)$: Lean `collision0, collision1, collisionValue, collision` and `collision_points_distinct` (`Counterexample.lean`, lines 143–170); Rocq `integral_collision_0_value, integral_collision_1_value`

and `integral_collision_points_distinct` (`Counterexample.v`, lines 249–265); and the refutation `jacobianConjectureInDimensionThree_false` (`Counterexample.lean`, line 201; Rocq `jacobian_conjecture_dimension_three_is_false`, `Counterexample.v`, line 331).

The weighted torus action on which the whole class rests is `counterexample_scaling` (`Algebra/JacobianConjecture/Lean/JacobianConjecture/Scaling.lean`, line 35; Rocq `Scaling.v`, line 32). The same group also checks the normalization (3.4), which is *not* formalized, and nothing else of Part II is: not the weighted Jacobian identity, not the normal forms (T) and (N), and none of the theorems, propositions or corollaries of Sections 15–21. The package contains no Lean or Rocq file. Placement of this report beside the Lean/Rocq development confers no formal status on it; the manuscript says so correctly (its source ledger: the package “neither edits the repository nor inherits the formal status of declarations stored elsewhere in it”).

13 The question, the contribution, and its boundaries

13.1 A specific unresolved question in the repository

ProveIt’s Jacobian project records the polynomial map F_0 of (1.1). The project attributes the map to Alpöge’s July 2026 announcement and contains Lean and Rocq developments for its constant Jacobian and explicit collisions [4]. The identities needed here are elementary: $\det DF_0 = -2$ and the collision (1.2). We check them independently in the companion code. They are not new results of this article. [write] The map and these two identities are printed once, in Part I; their kernel-checked status is in Section 12.5.

A subsequent repository report, *All-Degree Rigidity of a Weighted Keller Class* (Part I), classifies a normalized weighted class when the third invariant numerator is $r = R(t) + \gamma v$ with $\gamma \in K$. Its Section 10.2 explicitly asks whether allowing

$$r = R(t) + G(t)v$$

with a nonconstant polynomial G produces new normal forms, or whether the Keller and polynomiality conditions force G to be constant. The obstacle identified there is real: the substitution $s = r$ divides by G and is not a global polynomial coordinate change.

Theorem 15.1 resolves that question under the report’s retained hypotheses on p and q : *must be constant*. No bound on the ordinary degree is added. The proof explicitly controls the zeros of G ; it never treats the rational substitution as a polynomial automorphism.

13.2 What is new here, and what is inherited

The new field-level result is constant-slope rigidity for the full three-numerator affine-in- v class. The new infinitesimal results are the reduced-base extension, nonconstant dual-number examples, universal second-order obstruction at the simplest tame point, and the resulting nilpotent directions in finite coefficient schemes.

The constant-slope normal forms, their tame inverse, diagonal equivalence, source-shear extension, core support profile, and degree spectrum already occur in Part I. We rederive them after proving the missing rigidity step, both to make this article self-contained and to extend their *scope* to every affine-in- z map with the specified weights. The weighted determinant identity and its localization proof were already present in the project’s research notes, as acknowledged by that report. The cubic function-field calculation in Section 19 is likewise

not advertised as new for F_0 ; its role is to make the geometric consequences of the enlarged classification explicit.

13.3 Relation to the current primary literature

Shaska’s revised paper [5] studies the graded structure of the counterexample, quotient coordinates, and coefficient schemes. Gao [6] develops a broader construction of noninjective Keller maps from curves, including maps of different geometric degrees. Those works supply important context for why a structural classification inside a specified graded class is useful; neither is needed as a black-box classification theorem in our proof.

The source check for this article used the repository pin (Section 12.1), the precise prior report and its stated exclusions, Shaska’s version 2, and Gao’s version 1. This is a targeted comparison, not an exhaustive bibliographic priority determination. In particular, the present work is *not* a new disproof of the Jacobian conjecture, a classification of arbitrary Keller maps, or a minimum-degree theorem without symmetry and linearity hypotheses. “Degree” below always means ordinary polynomial degree unless explicitly qualified as function-field or geometric degree.

14 Weighted coordinates and the exact coefficient equations

Throughout Sections 14–21, K is a field of characteristic zero. We write primes for differentiation with respect to t .

Definition 14.1. A map $F = (P, Q, R) : \mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ is *weighted* in this article if

$$F(\rho^{-1}x, \rho y, \rho^2 z) = (\rho^2 P(x, y, z), \rho Q(x, y, z), \rho^{-1} R(x, y, z)) \quad (14.1)$$

as an identity with an invertible indeterminate ρ . It is *Keller* if $\det DF \in K^\times$.

Set $t = xy$ and $v = x^2 z$. A monomial $x^i y^j z^k$ of weight 2 satisfies $i = j + 2k - 2$, so multiplying it by x^2 gives $t^j v^k$. The corresponding exponents for weights 1 and -1 are $j + 2k - 1$ and $j + 2k + 1$. Consequently every map satisfying (14.1) has a unique expression $F = \mathcal{L}(p, q, r)$ in the localization at x , as in (2.2). It is affine in z in each coordinate if and only if

$$p = a(t)v + b(t), \quad q = c(t)v + d(t), \quad r = g(t)v + e(t) \quad (14.2)$$

with all six coefficients in $K[t]$. The lift is polynomial exactly when

$$b \in t^2 K[t], \quad d \in t K[t]. \quad (14.3)$$

Indeed, a monomial $t^i v^j$ in p or q must satisfy respectively $i + 2j \geq 2$ or $i + 2j \geq 1$; for $j \leq 1$ these are precisely (14.3). There is no analogous restriction on e or g .

For every polynomial lift, $\det DF = \mathcal{J}(p, q, r)$ with $\mathcal{J}$ the weighted determinant (2.4) evaluated at $(t, v) = (xy, x^2 z)$ (Proposition 2.2). [write] The manuscript states and proves this again as its Proposition 2.2, by the same chain-rule computation in the localization at x ; it is printed once, in Part I, and is the project’s (Section 1.4).

The substitution $K[t, v] \rightarrow K[x, y, z]$ is injective: distinct $t^i v^j$ produce distinct monomials $x^{i+2j} y^i z^j$. Thus we may work with $\mathcal{J}$ as a polynomial in t, v . Its value at $t = v = 0$ is

$$\mathcal{J}(0, 0) = -a(0)d'(0)e(0). \quad (14.4)$$

For a Keller map, all three factors are nonzero. Dividing the first, second, and third target coordinates respectively by these factors normalizes

$$a(0) = d'(0) = e(0) = 1, \quad \mathcal{J} = -1. \quad (14.5)$$

This normalization loses no Keller map and preserves the structural class.

For reference, direct expansion gives

$$\mathcal{J} = J_0 + J_1v + J_2v^2, \quad (14.6)$$

where

$$J_0 = (ce + dg)b' + 2b(ce' - gd') - a(de)', \quad (14.7)$$

$$J_1 = (ce + dg)a' + 2b(cg' - gc') + a(ce' - dg' - ec' - 3gd') + 2cgb', \quad (14.8)$$

$$J_2 = 2cga' - 3agc' + acg'. \quad (14.9)$$

These identities will also make the deformation calculation transparent.

15 The main classification

Theorem 15.1 (Constant-slope rigidity). *Let (14.2) satisfy polynomiality (14.3) and the normalized Keller identities (14.5). Then g is constant. Equivalently, every weighted Keller map affine in z has a constant third invariant slope r_v , after or before target normalization.*

Theorem 15.2 (Complete normal forms). *Under the hypotheses of Theorem 15.1, exactly one of the following two cases occurs.*

(T) Tame branch. *There are unique $\lambda \in K$ and $B \in t^2K[t]$ such that*

$$p = v + B(t), \quad q = t + \lambda(v + B(t)), \quad r = 1. \quad (15.1)$$

(N) Noninjective branch. *There are unique $\beta, \gamma \in K^\times$ and $e \in K[t]$ with $e(0) = 1$ and $e'(0) = \beta$ such that, on setting*

$$U = 1 - \frac{2\beta}{3}t, \quad S = e(t) + \gamma v, \quad (15.2)$$

one has

$$p = \frac{U^3S - (U^2 + U)/2}{\gamma}, \quad (15.3)$$

$$q = \frac{9}{\beta} \left(U^2S - \frac{2U + 1}{3} \right), \quad (15.4)$$

$$r = S. \quad (15.5)$$

Conversely every displayed triple gives a normalized polynomial Keller map. The maps in (T) are tame automorphisms, and those in (N) are noninjective over the ground field K .

Here no ordinary degree bound is imposed on a, b, c, d, e, g . The classification is therefore not an extrapolation from finitely many coefficient schemes. The parameters of (N) are read directly from r : $\gamma = r_v$, $e = r(t, 0)$, and $\beta = e'(0)$. The inverse and collision assertions are proved in Section 18.

Remark 15.3 ([write] Relation to Part I). Branch (T) is Part I's case (A), equation (3.2), verbatim. For $e = 1 + \beta t$, branch (N) is Part I's F_{a_I, b_I} of (3.1) with $a_I = \beta^2/\gamma$, $b_I = \gamma/\beta$: U is Part I's E and (15.3)–(15.4) expand to (5.11) before the γ -rescaling, since $-(U^2 + U)/2 = -1 + \beta t - \frac{2\beta^2}{9}t^2$ and $-\frac{9}{\beta} \cdot \frac{2U+1}{3} = 4t - \frac{9}{\beta}$. For general e it is Part I's $F_{a_I, b_I} \circ T_h$ of Theorem 6.3 (see (18.2)). What is new is that Theorem 15.1 makes these the *only* forms once r is merely affine in v . The normal form (N) with symbolic β, γ, e was rechecked at the write (Jacobian -1 identically).

16 Proof of constant-slope rigidity

The proof separates three issues that a formal coefficient search would not settle: arbitrary degree cancellations, poles of the moving coordinate, and polynomiality across the coordinate hyperplane.

16.1 Two elementary rational-function facts

Lemma 16.1 (Constants of differentiation). *If $f \in K(t)$ and $f' = 0$, then $f \in K$.*

Proof. Write $f = A/B$ with coprime polynomials. The identity $A'B - AB' = 0$ implies $A \mid A'$ if $A \neq 0$ and $B \mid B'$. A nonconstant polynomial in characteristic zero has derivative of strictly smaller degree, so both must be constant. The zero function is immediate. $\square$

Lemma 16.2 (A rational pole equation). *Let K be algebraically closed of characteristic zero. Suppose $U \in K[t] \setminus \{0\}$, $L \in K(t)$, and $\alpha \in K^\times$ satisfy*

$$LL' = \frac{\alpha}{U^3}. \quad (16.1)$$

Then for some $\xi \in K$ and nonzero scalars u, ℓ ,

$$U = u(t - \xi)^m, \quad L = \ell(t - \xi)^{-n}, \quad m, n \geq 1, \quad 3m = 2n + 1.$$

If, in addition, $H \in K[t]$ is coprime to U and $\frac{U}{H}(\delta_0 - L)$ is regular at ξ for a scalar δ_0 , then $m = n = 1$.

Proof. The right side is nonzero, so L and L' are nonzero. If L had a finite zero of order $j \geq 1$, the order of LL' there would be $2j - 1 > 0$. The right side of (16.1) has order either 0 or a negative multiple of 3. This is impossible. A rational function over an algebraically closed field with no finite zero has constant numerator in reduced form. Thus $L = \ell_0/V$ for a nonzero scalar ℓ_0 and a nonconstant polynomial V .

Substitution gives

$$U^3 V' = -\frac{\alpha}{\ell_0^2} V^3. \quad (16.2)$$

Every root of V' is therefore a root of V . If V has degree N and r distinct roots, the total multiplicity of those roots in V' is $N - r$: a root of multiplicity j contributes exactly $j - 1$ in characteristic zero. Since all $N - 1$ roots of V' occur there, $r = 1$. Write $V = c(t - \xi)^n$. Equation (16.2) then forces $U = u(t - \xi)^m$ and $3m = 2n + 1$. In particular $m, n \geq 1$.

Since $H(\xi) \neq 0$ and L has a pole, the order at ξ of $(U/H)(\delta_0 - L)$ is exactly $m - n = (1 - m)/2$. Regularity forces $m \leq 1$, so $m = n = 1$. $\square$

[write] Inside this proof only, r counts distinct roots and c is a scalar; neither is the numerator r or the coefficient polynomial c .

16.2 The nonzero-slope case has nonzero second slope

Assume $g \neq 0$. If $c = 0$, the equation $J_1 = 0$ reduces to

$$dga' - adg' - 3agd' = 0.$$

Here a, d, g are nonzero: $a(0) = 1$ and $d'(0) = 1$. Dividing in $K(t)$ shows that

$$\left(\frac{a}{gd^3}\right)' = 0.$$

Lemma 16.1 gives $a = \kappa gd^3$ with $\kappa \in K^\times$. But $d(0) = 0$, contradicting $a(0) = 1$. Hence

$$a \neq 0, \quad c \neq 0, \quad g \neq 0. \quad (16.3)$$

16.3 Cube-square factorization with coprime factors

We may extend scalars to an algebraic closure for this part of the proof. A polynomial proved constant after this extension was already constant over the original field. Equation $J_2 = 0$ gives

$$\left(\frac{a^2g}{c^3}\right)' = 0, \quad \frac{a^2g}{c^3} \in K^\times$$

by Lemma 16.1. At a finite point, write $o_a = \text{ord}(a)$, $o_c = \text{ord}(c)$, $o_g = \text{ord}(g)$. Then

$$2o_a + o_g = 3o_c. \quad (16.4)$$

The three polynomials a, c, g cannot vanish at the same point: otherwise the last column of the determinant (2.4) would vanish there for every v , contradicting $\mathcal{J} = -1$.

If $o_g > 0$, equation (16.4) and the preceding observation force $o_a = 0$ and $o_g = 3o_c$. If $o_a > 0$, then $o_g = 0$ and $2o_a = 3o_c$, so o_a is a multiple of 3. Unique factorization, including nonzero scalar factors, therefore gives

$$a = U^3, \quad c = kHU^2, \quad g = H^3, \quad (U, H) = 1, \quad (16.5)$$

for polynomials U, H and a scalar $k \neq 0$. We choose the cube root in U so that $U(0) = 1$, which is possible because $a(0) = 1$. Choosing a cube root of the scalar in g is harmless over the algebraic closure. Scalar descent will be automatic once g is proved constant.

16.4 A rational moving coordinate and its differential invariant

Work temporarily over $K(t)$, and set

$$w = \frac{U}{H}, \quad s = e + H^3v.$$

Then uniquely

$$p = w^3s + B, \quad q = kw^2s + D, \quad r = s, \quad (16.6)$$

where

$$B = b - w^3e, \quad D = d - kw^2e.$$

The substitution $(t, v) \mapsto (t, s)$ has Jacobian H^3 . Accordingly the old determinant equals H^3 times the weighted determinant computed in (t, s) . For arbitrary A, B, C, D , the latter is

$$\begin{aligned} &\mathcal{J}(As + B, Cs + D, s) \\ &= DB' - 2BD' + s(DA' + 2CB' - 3AD' - 2BC') + s^2(2CA' - 3AC'). \end{aligned} \quad (16.7)$$

This identity is just a determinant expansion; the change of variables is being used only rationally. [write] It is Part I's (5.3), whose coefficients are the left sides of (4.4)–(4.6); there $A, \dots, D$ are polynomials, here rational functions.

Put $A = w^3$, $C = kw^2$, and then write

$$B = w^2T, \quad D = wV.$$

The s^2 coefficient vanishes identically. The s coefficient becomes

$$w^4(2kT' - 3V').$$

It follows that

$$V = \alpha T + \delta, \quad \alpha = \frac{2k}{3} \neq 0, \quad \delta \in K. \quad (16.8)$$

The constant coefficient satisfies

$$DB' - 2BD' = w^3(VT' - 2TV') = w^3(\delta - \alpha T)T'.$$

Multiplying by H^3 and using $\mathcal{J} = -1$ gives

$$-1 = U^3(\delta - \alpha T)T'. \quad (16.9)$$

Therefore

$$L := \delta - \alpha T \quad \text{satisfies} \quad LL' = \frac{\alpha}{U^3}.$$

16.5 Regularity collapses every higher-degree pole pattern

Lemma 16.2 applies. Its additional regularity condition also holds:

$$D = d - kw^2e = w(2\delta - L)$$

is regular at any root of U , because d, e are polynomials and H is coprime to U . Thus U is linear and L is a nonzero scalar multiple of $1/U$. Rewriting the constants, we obtain

$$T = \eta + \frac{\theta}{U}, \quad \delta = \alpha\eta, \quad \alpha\theta^2U' = -1. \quad (16.10)$$

In particular $U' \neq 0$ and $\theta \neq 0$.

This is the all-degree step: the initial numerator polynomials could have arbitrarily large degree, but the pole equation and regularity force a single simple pole and a single linear common factor.

16.6 Polynomiality at the origin forces the remaining slope constant

Recovering the original constant terms from (16.6)–(16.10) gives

$$b = \frac{U^3e + H(\eta U^2 + \theta U)}{H^3}, \quad (16.11)$$

$$d = \frac{kU^2e + (k/3)H(4\eta U + 2\theta)}{H^2}. \quad (16.12)$$

Since b, d are polynomials and $(U, H) = 1$, these identities imply

$$\begin{aligned} H^3 & \mid U^2e + H(\eta U + \theta), \\ H^2 & \mid U^2e + \frac{H}{3}(4\eta U + 2\theta). \end{aligned}$$

Subtracting modulo H^2 yields

$$H \mid \eta U - \theta. \quad (16.13)$$

We next show $H(0) \neq 0$. Multiplying (16.11) by H^3 and evaluating at 0, a hypothetical $H(0) = 0$ would give $0 = U(0)^3 e(0) = 1$. Set $h_0 = H(0) \neq 0$. The conditions $b(0) = d(0) = 0$, $U(0) = e(0) = 1$ now give

$$1 + h_0(\eta + \theta) = 0, \quad 1 + \frac{h_0}{3}(4\eta + 2\theta) = 0.$$

Subtracting three times the first equation from the second after clearing its denominator shows $\eta = \theta$. Both are nonzero. Therefore (16.13) says $H \mid U - 1$. Since U is linear with $U(0) = 1$, $U - 1$ is a nonzero scalar multiple of t . But $H(0) \neq 0$, so H is constant. Thus $g = H^3$ is constant.

This proves Theorem 15.1 when $g \neq 0$; the case $g = 0$ was already constant. The algebraic-closure argument proves vanishing of all positive-degree coefficients of the original g , so it descends to every characteristic-zero field. $\square$

17 Reconstruction and the exceptional branch

17.1 Zero slope forces the tame form

Suppose $g = 0$. Equation $J_1 = 0$ becomes

$$e'ac + e(a'c - ac') = 0.$$

If $c \neq 0$, Lemma 16.1 applied to $c/(ea)$ shows $c = \lambda ea$ for a scalar λ . The same expression holds for $c = 0$ with $\lambda = 0$. The constant coefficient equation becomes

$$a(\lambda e^2 b - ed)' = -1. \quad (17.1)$$

The two factors are polynomials, so both are constant. Since $a(0) = 1$, $a = 1$. Integrating and using $b(0) = d(0) = 0$ gives

$$\lambda e^2 b - ed = -t.$$

Consequently $e \mid t$. The condition $e(0) = 1$ forces $e = 1$, whence $c = \lambda$ and $d = t + \lambda b$. Together with $b \in t^2 K[t]$, this is exactly (15.1). [write] This is the argument of Part I's Theorem 6.3, case $\gamma = 0$, in the present notation.

17.2 Nonzero slope gives the explicit noninjective family

Let $g = \gamma \neq 0$. The replacement

$$\tilde{p}(t, v) = \gamma p(t, v/\gamma), \quad \tilde{q}(t, v) = q(t, v/\gamma), \quad \tilde{r}(t, v) = r(t, v/\gamma) \quad (17.2)$$

preserves polynomiality, the normalized origin coefficients, and the Jacobian -1 . It reduces the third numerator to $e + v$.

We may therefore use the proof above with $H = 1$. The two origin equations now give

$$\eta = \theta = -\frac{1}{2}.$$

Put $\beta = e'(0)$. Differentiating (16.11) at zero and using $b'(0) = 0$ gives

$$0 = \frac{3}{2}U' + \beta.$$

Thus $U = 1 - 2\beta t/3$ and $\beta \neq 0$, since $U' \neq 0$. The last identity in (16.10) gives $kU' = -6$, hence $k = 9/\beta$. We recover

$$\tilde{p} = U^3(e + v) - \frac{U^2 + U}{2}, \quad \tilde{q} = \frac{9}{\beta} \left(U^2(e + v) - \frac{2U + 1}{3} \right).$$

Undoing (17.2) yields (15.3)–(15.5). Every coefficient is expressed over the original field, so no algebraic-extension parameter remains. [write] This is a second proof of Part I's nonconstant branch (Section 5, case $\gamma \neq 0$), and of Theorem 6.3(ii) without the source shear: Part I bounds degrees by the weighted Wronskian lemmas, the manuscript by the pole equation.

17.3 Converse and uniqueness

For (15.1), direct substitution gives $\mathcal{J} = -1$. For (15.3)–(15.5), first set $\gamma = 1$ as above. Then

$$A = U^3, \quad C = \frac{9}{\beta}U^2, \quad B = -\frac{U^2 + U}{2}, \quad D = -\frac{3}{\beta}(2U + 1).$$

Equation (16.7) has zero s^2 and s coefficients, and constant coefficient -1 , using $U' = -2\beta/3$. This argument is independent of e . Restoring γ preserves the determinant.

At $t = 0$ the formulas give $b(0) = d(0) = 0$. Differentiation gives $b'(0) = 0$, $d'(0) = 1$, and $a(0) = 1$, while $e(0) = 1$ by hypothesis. Thus the lift is polynomial and normalized. In (T), the parameters are $B = p(t, 0)$ and $\lambda = q_v$; in (N), they are read from r as stated above. The branches are disjoint because $r_v = 0$ in (T) and $r_v \neq 0$ in (N). This proves the algebraic assertions of Theorem 15.2.

18 Equivalence, certificates, and exact degree consequences

18.1 An explicit inverse on the tame branch

Write $B = t^2H$. The tame lift of (15.1) is (6.1), and its inverse is (6.2). Substitution in either order gives the identity. Moreover the map is the composition of the elementary shear $z \mapsto z + y^2H(xy)$, the elementary shear $y \mapsto y + \lambda xz$ in the updated coordinates, and the permutation $(x, y, z) \mapsto (z, y, x)$. Hence it is tame, not merely invertible. [write] The manuscript displays the lift and the inverse again, with output coordinates (X, Y, Z) where Part I writes (U, V, W) ; they are printed once, in Part I. Here H is Part I's H with $B = t^2H$, not the cube root of Section 16.

18.2 One core and one source shear on the noninjective branch

Let $F_{\beta, \gamma}^{\text{core}}$ denote the form (N) with $e = 1 + \beta t$, and let $F_* = F_{1,1}^{\text{core}}$. Define

$$h(t) = \frac{e(t) - 1 - \beta t}{\gamma t^2}, \quad T_h(x, y, z) = (x, y, z + y^2h(xy)). \quad (18.1)$$

The numerator defining h vanishes to order at least two, so $h \in K[t]$. The map T_h is an equivariant elementary shear with inverse T_{-h} , and on invariants it replaces v by $v + t^2h(t)$. Therefore

$$F = F_{\beta, \gamma}^{\text{core}} \circ T_h. \quad (18.2)$$

The core itself satisfies

$$F_{\beta, \gamma}^{\text{core}} = \text{diag}(\gamma^{-1}, \beta^{-1}, 1) \circ F_* \circ \text{diag}(1, \beta, \gamma), \quad (18.3)$$

and F_* is the normalization (3.4) of F_0 . These are exact polynomial identities. They also explain the relationship with the previous report's parameters a_I, b_I : there $\beta = a_I b_I$ and $\gamma = a_I b_I^2$, so $a_I = \beta^2/\gamma$ and $b_I = \gamma/\beta$. We avoid using a, b as family parameters here because they already name the first numerator coefficients in (14.2). [write] (18.1) is Part I's h of (6.9) with $R = e$, (18.2) is Theorem 6.3(ii), and (18.3) is (6.3) with $F_{\beta, \gamma}^{\text{core}} = F_{a_I, b_I}$.

Corollary 18.1 (Two tame source–target equivalence classes). *Every map in the stated weighted affine-in- z Keller class is tame source–target equivalent either to the identity or to F_0 , and the two classes are distinct.*

Proof. Undo target normalization. The tame branch consists of tame automorphisms, so lies in the identity class. Equations (18.2), (18.3) and (3.4) put the other branch in the class of F_0 . Noninjectivity, shown next, is preserved by source and target automorphisms and separates the two classes. $\square$

18.3 Ground-field collision certificates

The core has the collision

$$F_{\beta, \gamma}^{\text{core}}(1, 0, -1/\gamma) = F_{\beta, \gamma}^{\text{core}}(0, -9/\beta, 71/\gamma) = (-1/\gamma, -9/\beta, 0).$$

Transporting by T_{-h} gives the following certificate for the full class.

Proposition 18.2. *For every normalized map in (N) , the distinct points*

$$\begin{aligned} X_1 &= (1, 0, -1/\gamma), \\ X_2 &= \left(0, -9/\beta, \frac{71}{\gamma} - \frac{81h(0)}{\beta^2}\right) \end{aligned} \tag{18.4}$$

have common image $(-1/\gamma, -9/\beta, 0)$.

Proof. At $t = 0$, the invariant numerators are $p = v$, $q = 9\gamma v/\beta$, and $r = 1 + \gamma v$. This gives the image of X_1 directly. On $x = 0$, the core is

$$(0, y, z) \mapsto \left(z - \frac{8\beta^2}{9\gamma}y^2, y, 0\right).$$

Putting $y = -9/\beta$, $z = 71/\gamma$ gives the same image. The shear does not move X_1 and sends X_2 to that core point. The first coordinates of X_1 and X_2 differ, so they are distinct over every characteristic-zero field. $\square$

[write] This is Part I's collision (6.6) (with $\gamma = a_I b_I^2$, $\ell = 9b_I = 9\gamma/\beta$), made explicit after the shear that Part I's proof of Theorem 6.3 only mentions; the mechanism is the project's (Section 1.4). The proposition was rechecked at the write with symbolic β, γ and $e = 1 + \beta t + e_2 t^2 + e_3 t^3$.

Unnormalizing the target changes the common output by a diagonal scaling, but not the colliding source points. Thus the classifier supplies rational collision certificates whenever the input coefficients are rational.

Example 18.3. Take $\beta = \gamma = 1$ and $e = 1 + t + t^2 + t^3$. Then $h = 1 + t$, and the collision is

$$F(1, 0, -1) = F(0, -9, -10) = (-1, -9, 0).$$

Its coordinate degrees are $(10, 9, 7)$, even though its underlying core has degrees $(7, 6, 4)$. The additional degree comes entirely from the source shear, not from a new equivalence class.

18.4 The exact ordinary degree spectrum

Corollary 18.4. *For every normalized noninjective map in this class, $h = 0$ gives coordinate degrees $(7, 6, 4)$ and core support sizes $(7, 6, 3)$. If $h \neq 0$ has degree m , the coordinate degrees are exactly*

$$(2m + 8, 2m + 7, 2m + 5). \quad (18.5)$$

Consequently the possible maximum ordinary degrees of noninjective maps in the full weighted affine-in- z class are exactly

$$\{7\} \cup \{8, 10, 12, 14, \dots\}.$$

Proof. [write] By Theorem 15.1 and (18.2), every such map is one of Part I's Theorem 6.3, so this is Corollary 6.4 together with Corollary 6.1. The manuscript's proof repeats Part I's: expansion of the core formulas gives the degree and support profiles ($\beta\gamma \neq 0$ makes every listed core coefficient nonzero); the coefficients of z in the three coordinates have degrees 6, 5, 3; substituting $z + y^2h(xy)$ adds nonzero leading terms of degrees $6 + (2 + 2m)$, $5 + (2 + 2m)$, $3 + (2 + 2m)$, which no old term can cancel; and $h = t^m$ realizes every $m \geq 0$. $\square$

This is a structural exclusion of ordinary degrees 4, 5, 6 for *noninjective maps in this class*, not for all three-variable Keller maps. Likewise the sixteen-term assertion concerns the unsheared core; we make no unsupported claim that arbitrary shears preserve total support size or that sixteen terms are globally minimal.

19 Function-field consequences and a cubic model

The constant-slope core and its cubic geometry were already part of the post-announcement discussion reflected in [5, 6]. The enlarged classification now transfers that geometry to every noninjective map of our class. For completeness we give a direct calculation for F_0 .

Proposition 19.1. *Let $(P, Q, R) = F_0(x, y, z)$. Then*

$$[K(x, y, z) : K(P, Q, R)] = 3,$$

and the Galois group of the normal closure is S_3 . The same statements hold after extending K to its algebraic closure, and for every map in branch (N). In branch (T) the degree is 1.

Proof. The nonzero Jacobian implies algebraic independence of P, Q, R in characteristic zero. To see this directly, choose a nonzero polynomial relation of least positive degree if one exists. Differentiating it and inverting the Jacobian over $K(x, y, z)$ shows that every partial derivative of the relation is another relation. Minimality makes all those partial derivatives zero; characteristic zero then makes the relation constant, a contradiction.

Set $\tau = y + 1/x$. Expansion of (1.1) gives

$$f(\tau) = 0, \quad f(T) = RT^3 - 2T^2 + QT - 2P, \quad (19.1)$$

and

$$\frac{1}{x} = \frac{Q - 4\tau + 3R\tau^2}{2} = \frac{f'(\tau)}{2}, \quad y = \tau - \frac{1}{x}, \quad z = \frac{2x - 3x^2y - R}{x^3}. \quad (19.2)$$

Thus $K(x, y, z) = K(P, Q, R, \tau)$.

As a polynomial in $K[P, Q, R, T]$, f is irreducible: it is linear in P with invertible coefficient -2 , and its quotient ring is the polynomial ring $K[Q, R, T]$. Viewed as a polynomial in T

over $K[P, Q, R]$, it is primitive because its T^2 coefficient is -2 . Gauss's lemma therefore gives irreducibility in $K(P, Q, R)[T]$. The extension degree is three.

Its discriminant is

$$\Delta = 4(Q^2 - RQ^3 - 16P - 27R^2P^2 + 18PQR). \quad (19.3)$$

It is not a square in $K(P, Q, R)$. Indeed, a polynomial that is a rational square over a unique factorization domain is a polynomial square: a coprime denominator would have to divide the numerator. But specializing such a square identity at $R = 0$ would make $4(Q^2 - 16P)$ a square in $K[P, Q]$, impossible because its degree in P is one.

An irreducible separable cubic has a transitive subgroup of S_3 as its normal-closure group. Its discriminant is a square exactly when the group lies in A_3 : the product of the three pairwise root differences changes by the sign of a permutation. The nonsquare discriminant therefore forces S_3 . The argument is unchanged over the algebraic closure of K . Source and target polynomial automorphisms preserve these function-field invariants, so (18.2), (18.3) and (3.4) transfer them to (N). The explicit inverse gives degree one in (T). $\square$

This statement concerns generic function-field degree. It does not assert that every special fiber has three points, that the map is finite, or that its restriction over an unspecified open set is already a covering.

[write] The cubic is not new in ProveIt either. The sibling report `jacobian-conjecture/arithmic-local-global-fibers` uses the same variable τ (its $t = y + 1/x$), the same cubic $g_{A,B,C}(T) = CT^3 - 2T^2 + BT - 2A$ with $(A, B, C) = (P, Q, R)$, the same derivative identity $f'(\tau) = 2/x$ and the same discriminant (its Section 2, equation (2.9), equal to (19.3) after expansion); it credits the cubic inverse model to an archived note of `royvanrijn/jacobian-research` and the complex fiber law to Gao's Theorems 3.3–3.4, and Gao's Theorem 3.3 gives geometric degree three for this map (Section 1.5). The sibling report does not state that the generic Galois group is S_3 . The cubic relation, the discriminant identity and the degree profile of Example 18.3 were rechecked at the write.

20 Reduced bases and nilpotent slope directions

Field-valued classification and scheme-theoretic classification are not interchangeable. This section determines an explicit part of the difference, rather than assuming that the field theorem automatically eliminates all coefficient directions.

20.1 Constancy over every reduced base

Theorem 20.1 (Reduced-base rigidity). *Let Λ be a reduced commutative $\mathbb{Q}$ -algebra with identity. Suppose $a, b, c, d, e, g \in \Lambda[t]$ satisfy*

$$b(0) = b'(0) = d(0) = 0, \quad a(0) = d'(0) = e(0) = 1, \quad \mathcal{J} = -1.$$

Then $g \in \Lambda$: every positive-degree coefficient of g vanishes.

Proof. For each prime ideal $\mathfrak{p}$ of Λ , reduce the identities modulo $\mathfrak{p}$ and pass to $\text{Frac}(\Lambda/\mathfrak{p})$, a field of characteristic zero. Theorem 15.1 makes every positive-degree coefficient of g zero there, hence zero in the domain $\Lambda/\mathfrak{p}$. Thus each such coefficient belongs to every prime ideal of Λ , that is, to its nilradical. Reducedness makes it zero. $\square$

We assert only slope constancy over reduced bases, not a global two-branch normal-form theorem for arbitrary rings. For example, a constant slope can be a zero divisor on a disconnected base, and division by it need not be allowed. [write] Part I's Corollary 7.2 is the same argument for $r = 1 + t + v$ fixed; here r is free.

20.2 Explicit dual-number solutions in every slope degree

Let $\varepsilon^2 = 0$. For $G \in K[t]$, define

$$C_G(t) = -tG(t) - 2 \int_0^t G(s) ds. \quad (20.1)$$

The integral here is formal polynomial antiderivation in characteristic zero; in particular $C_G(0) = 0$ and

$$C'_G = -tG' - 3G. \quad (20.2)$$

Theorem 20.2 (Nonconstant infinitesimal slopes). *For every $G \in K[t]$, the triple over $K[\varepsilon]/(\varepsilon^2)$*

$$p = v, \quad q = t + \varepsilon C_G(t)v, \quad r = 1 + \varepsilon G(t)v \quad (20.3)$$

is polynomial, normalized, and satisfies $\mathcal{J} = -1$. If G is nonconstant, its third invariant slope is a nonconstant polynomial over this nonreduced coefficient ring.

Proof. Polynomiality and the origin conditions are immediate. Over the untruncated polynomial ring $K[\varepsilon, t, v]$, formulas (14.7)–(14.9) give the exact identity

$$\mathcal{J} = -1 + \varepsilon^2 v^2 \mathcal{O}(G), \quad \mathcal{O}(G) = C_G G' - 3G C'_G. \quad (20.4)$$

The first-order term vanishes by (20.2). Modulo ε^2 , the displayed determinant is -1 . $\square$

For the monomial $G = t^m$ with $m \geq 0$, these formulas read

$$C_G = -\frac{m+3}{m+1} t^{m+1}, \quad (20.5)$$

$$\mathcal{O}(G) = \frac{(m+3)(2m+3)}{m+1} t^{2m}. \quad (20.6)$$

When $m \geq 1$, the third slope is genuinely nonconstant. In source coordinates the dual-number map is

$$(z, y + \varepsilon x z C_G(xy), x + \varepsilon x^3 z G(xy)). \quad (20.7)$$

For $G = t^m$, its maximum ordinary degree is $2m + 4$.

20.3 Every nonzero slope direction is obstructed at second order

One might try to repair (20.4) by adding second-order terms. The next theorem rules out *all* such repairs inside the structural class, not merely the obvious ones.

Theorem 20.3 (Universal second-order slope obstruction). *Start from the normalized tame point $(p_0, q_0, r_0) = (v, t, 1)$, whose lift is (z, y, x) . Consider any normalized first-order deformation, affine in v ,*

$$\begin{aligned} p &= v + \varepsilon(A(t)v + B(t)), \\ q &= t + \varepsilon(C(t)v + D(t)), \\ r &= 1 + \varepsilon(E(t) + G(t)v) \end{aligned} \quad (20.8)$$

that satisfies $\mathcal{J} = -1$ modulo ε^2 . If $G \neq 0$, this deformation has no lift to a Keller solution over $K[\varepsilon]/(\varepsilon^3)$ with all three numerators still affine in v . This remains true with no bound on the t -degrees of the second-order correction polynomials.

Proof. Linearizing (14.7)–(14.9) gives

$$[\varepsilon]J_0 = -(A + D' + E + tE'), \quad (20.9)$$

$$[\varepsilon]J_1 = -(C' + tG' + 3G), \quad (20.10)$$

$$[\varepsilon]J_2 = 0.$$

Thus every first-order solution satisfies

$$C' = -tG' - 3G. \quad (20.11)$$

If G has degree $m \geq 0$ and leading coefficient $g_m \neq 0$, then C has degree $m + 1$ and leading coefficient

$$-\frac{m+3}{m+1}g_m.$$

The integration constant in C is arbitrary and does not affect this statement.

Now add arbitrary affine-in- v second-order terms. In the notation of (14.2), write

$$a = 1 + \varepsilon A + \varepsilon^2 A_2, \quad c = \varepsilon C + \varepsilon^2 C_2, \quad g = \varepsilon G + \varepsilon^2 G_2,$$

with equally arbitrary b, d, e corrections. The coefficient J_2 depends only on a, c, g . Its formula (14.9) gives

$$[\varepsilon^2]J_2 = CG' - 3GC'. \quad (20.12)$$

In particular no second-order correction occurs in this expression, and neither do A, B, D, E .

The leading coefficient of (20.12), in degree $2m$, is

$$\frac{(m+3)(2m+3)}{m+1}g_m^2. \quad (20.13)$$

For $m = 0$, the same expression follows directly from $G' = 0$ and $C' = -3G$. Characteristic zero makes (20.13) nonzero. Therefore J_2 cannot vanish modulo ε^3 , as the Keller identity would require. This contradiction proves the theorem. $\square$

[write] In this theorem $A, \dots, E, G$ are first-order variations and g_m is the leading coefficient of G ; the coefficients g_i of Corollary 20.6 are those of the universal slope g .

The obstruction includes *constant* nonzero G , not just the nonconstant slopes that contradict reduced-base rigidity. It concerns first-order directions at the particular tame point (z, y, x) ; it does not say that the noninjective constant-slope branch is absent from the coefficient space. A branch can meet a point with its slope vanishing to higher order in a parameter.

20.4 The obstruction is sharp at the linearity boundary

The failure to lift is caused by the affine-in- v restriction, not by an intrinsic failure of the infinitesimal volume-preserving deformation. There is an explicit minimal-degree repair.

Proposition 20.4 (A quadratic repair). *For the canonical direction (20.3), set*

$$I_G(t) = \int_0^t \mathcal{O}(G)(s) ds.$$

Then, over $K[\varepsilon]/(\varepsilon^3)$,

$$p = v, \quad q = t + \varepsilon C_G v + \varepsilon^2 I_G v^2, \quad r = 1 + \varepsilon G v \quad (20.14)$$

is a normalized polynomial Keller map. For $G \neq 0$, maximum v -degree two is the smallest possible for a second-order repair of this direction.

Proof. At $(p, q, r) = (v, t, 1)$, the linearized weighted determinant for arbitrary polynomial variations (P_1, Q_1, R_1) is

$$-\partial_v P_1 - \partial_t Q_1 - R_1 - t \partial_t R_1 - 2v \partial_v R_1. \quad (20.15)$$

The second-order addition in (20.14) therefore contributes $-I'_G v^2 = -\mathcal{O}(G)v^2$, canceling (20.4) modulo ε^3 . The extra monomial powers lift polynomially and do not affect the normalized origin conditions. Theorem 20.3 rules out every degree-one repair when $G \neq 0$, whereas the displayed one has degree two. $\square$

[write] Theorem 20.2 with (20.5)–(20.6) and this repair were rechecked at the write for $G = t^m$, $0 \leq m \leq 4$.

Proposition 20.5 (Formal integration after removing the degree cap). *Every canonical direction (20.3) extends to a weighted formal Keller family over $K[x, y, z][[\varepsilon]]$. No uniform bound on the z -degree of its coefficients is asserted.*

Proof. Consider the polynomial derivation

$$\mathcal{D} = x^3 z G(xy) \frac{\partial}{\partial x} + x z C_G(xy) \frac{\partial}{\partial y}, \quad \mathcal{D}(z) = 0.$$

Its divergence is $x^2 z (3G + tG' + C'_G) = 0$. It preserves weights, and its coefficients have no linear terms. The formal exponential $\Phi_\varepsilon = \exp(\varepsilon \mathcal{D})$ is a well-defined automorphism of $K[x, y, z][[\varepsilon]]$, with inverse $\Phi_{-\varepsilon}$: every coefficient in ε is a polynomial, and the exponential identities hold coefficientwise. The formal flow preserves the Jacobian determinant. More explicitly, differentiating its Jacobian matrix in ε and applying the determinant product rule gives

$$\frac{\partial}{\partial \varepsilon} \det D\Phi_\varepsilon = (\operatorname{div} \mathcal{D}) \circ \Phi_\varepsilon \det D\Phi_\varepsilon = 0.$$

The initial determinant is one. Thus $F_\varepsilon = (z, y, x) \circ \Phi_\varepsilon$ has determinant -1 , preserves the required weights, and retains the normalized linear part. Its first-order term is exactly (20.7). This is the standard formal-flow construction applied to the present divergence-free direction, not a claim of a polynomial family in ε or analytic convergence. $\square$

20.5 Nonzero nilpotents in finite coefficient schemes

For $D \geq 2$, let Λ_D be the universal $\mathbb{Q}$ -algebra of coefficients of six polynomials a, b, c, d, e, g of t -degree at most D , modulo polynomiality, normalization, and the coefficient equations $\mathcal{J} = -1$. This is a finite presentation: only finitely many coefficients occur in (14.7)–(14.9). Write

$$g(t) = g_0 + g_1 t + \cdots + g_D t^D$$

for its universal third slope, and let $\mathbf{m}$ be the rational point corresponding to $(p, q, r) = (v, t, 1)$.

Corollary 20.6 (Arbitrarily many nilpotent slope directions). *For every $D \geq 2$:*

- (i) Every g_i with $i \geq 1$ is nilpotent in Λ_D .
- (ii) For $1 \leq i \leq D-1$, g_i is nonzero both in Λ_D and in the local ring $(\Lambda_D)_{\mathfrak{m}}$.
- (iii) The tangent space at $\mathfrak{m}$ contains a $(D-1)$ -dimensional subspace of nonconstant-slope directions, every nonzero member of which is obstructed from lifting to order two inside the affine-in- v class, even if the degree cap is subsequently removed.

Proof. Apply Theorem 20.1 to Λ_D modulo its nilradical to prove (i). For (ii), the dual-number example with $G = t^i$ uses C_G of degree $i+1$, so lies inside the degree cap when $i \leq D-1$. Its evaluation homomorphism $\Lambda_D \rightarrow \mathbb{Q}[\varepsilon]/(\varepsilon^2)$ sends g_i to $\varepsilon \neq 0$. Since the residue point is $\mathfrak{m}$, every element outside $\mathfrak{m}$ maps to a unit; the homomorphism extends to the local ring. Thus g_i is also nonzero there.

The first derivatives of these evaluation maps are tangent vectors whose values on the coordinates $g_1, \dots, g_{D-1}$ form the identity matrix. They are therefore linearly independent. A nonzero linear combination has a nonzero polynomial G with no constant term, and Theorem 20.3 applies. This proves (iii). $\square$

Thus a field-valued classification does not describe the full coefficient scheme: nonconstant slopes are invisible at all field-valued points but survive as genuine nilpotents. We have not determined the nilpotence indices, all relations among these nilpotents, or the full local algebra. Those are separate problems rather than consequences of the present calculation. [write] These are statements at the *tame* point $(v, t, 1)$ with r free. Part I's Theorem 8.1 fixes $r = 1 + t + v$ at the noninjective point; the two coefficient problems are different, and neither result bears on the other's scheme structure.

21 An exact recognition algorithm and its verification boundary

21.1 Recognition without a determinant search

Assume the input consists of six densely encoded univariate polynomials of maximum degree D over a characteristic-zero field with exact field operations. The following algorithm decides the Keller condition inside the stated structural class.

1. Check $b(0) = b'(0) = d(0) = 0$. A failure means the proposed lift is not a polynomial map in the class.
2. Compute $a(0)$, $d'(0)$, and $e(0)$. A zero factor implies that the Jacobian vanishes at the origin, so the map is not Keller. Otherwise normalize the three outputs by these factors.
3. Check whether g is constant. If not, reject by Theorem 15.1.
4. If $g = 0$, test $r = 1$, $p = v + B$, and $q = t + \lambda(v + B)$ with $B = p(t, 0)$ and $\lambda = q_v(0)$. On success, return the inverse (6.2) and undo output normalization.
5. If $g = \gamma \neq 0$, compute $\beta = e'(0)$. Reject if $\beta = 0$; otherwise compare the input with (15.3)–(15.5). On success return β, γ, e, h and the collision (18.4).

Theorem 15.2 proves correctness in both directions. No polynomial factorization, extraction of roots, algebraic closure, or unbounded search is required by the algorithm. Those tools are used in the proof of the theorem, not by an implementation of the normal-form test. [write]

This extends Part I's procedure of Section 9, whose scope stopped at a constant coefficient of v in r .

Proposition 21.1 (Arithmetic-operation complexity). *The normal-form recognition algorithm uses $O(D)$ field operations for dense univariate coefficient arrays, in addition to reading the input.*

Proof. All origin tests, scalar normalizations, derivative coefficient reads, and coefficient comparisons take linear time. The only polynomial multiplications needed in (N) are by U^2 and U^3 , of fixed degrees two and three. The shift defining h is a subtraction followed by scalar and monomial division. Each operation is linear in the input degree. $\square$

This is an arithmetic model bound, not a bit-complexity bound over $\mathbb{Q}$. Nor is it a performance guarantee for a general-purpose symbolic expression engine. The companion SymPy program implements the same tests exactly but uses expression expansion and polynomial conversion for clarity.

21.2 The executable companion

The package contains `code/weighted_keller.py`, restricted to rational input coefficients. Its public functions provide weighted Jacobians, polynomial lifting, normalized noninjective forms, classification, collision points, and infinitesimal obstruction polynomials. Inputs outside the structural class raise a descriptive `ValueError`; a valid polynomial input in the class that is not Keller returns a negative classification.

For example:

```
from code.weighted_keller import t, normal_form, classify
p, q, r = normal_form(1, 1, 1 + t + t**2 + t**3)
result = classify(p, q, r)
assert result.kind == "noninjective"
```

The executable implementation accepts rational coefficients rather than arbitrary abstract characteristic-zero fields. This restriction belongs to the software, not to the theorem. [write] In ProveIt the module is shipped as `code/02-affine-slope-weighted_keller.py`, which is not an importable module name; the example works in a copy with the delivered names (Section 25.2).

21.3 Twelve exact-check groups

The script `code/verify_results.py` was run with SymPy 1.14.0. It records its Python and SymPy versions and all results in `data/verification.json`, with a compact text summary alongside. Every group passed:

Group	Exact assertion checked
1	The three generic coefficients of the weighted determinant.
2	The rational moving-coordinate and pole-equation reductions.
3	The noninjective normal form with symbolic $\beta, \gamma, e(t)$.
4	$\det DF_0 = -2$, the integral collision, and the exact normalization.
5	Rational diagonal equivalence and source-shear examples.
6	Both tame inverse identities for a symbolic function H .
7	Eighteen noninjective rational cases, eighteen rejected mutations, twelve tame cases, and three exceptional rejections.
8	The core degree/support profiles and shears of degrees 0 through 10.
9	Generic first-order equations and independence of the second-order obstruction from all second-order affine-in- v corrections.
10	Dual-number examples $G = t^m$ for $0 \leq m \leq 12$ and three mixed polynomials; the quadratic second-order repair.
11	Cubic elimination, the inverse x formula, and the discriminant.
12	Input-domain, polynomiality, and singular-origin validation.

The randomized rational examples use the fixed seed 20260929. Generic symbolic identities are distinguished from finite examples in the JSON output. The root-multiplicity, pole, nilradical, and all-degree arguments are written proofs; finite verification is not substituted for them. [write] The paths in this subsection are the delivered ones; in ProveIt they are `code/02-affine-slope-verify_results.py` and `data/02-affine-slope-verification.json` (and `.txt`). Part I's own script is the unprefix `code/verify_results.py`. Group 4 rechecks kernel-checked statements (Section 12.5).

22 Further research questions and concrete next steps

The original nonconstant-slope question is settled by Theorem 15.1; it should not remain on the open-problem list under the same hypotheses. The following extensions remain distinct.

22.1 Quadratic dependence on the second invariant

Allow $\deg_v p, \deg_v q \leq 2$, initially retaining $\deg_v r \leq 1$. Must the third slope still be constant? If not, classify the smallest new families up to equivariant tame equivalence. The present pole argument uses both the linear moving-coordinate expansion and the exponent pattern in J_2 ; neither automatically survives an added v^2 coefficient. A useful first milestone is an all-degree description of the highest two coefficient equations, rather than a fixed ordinary-degree search alone. This is the next linearity extension proposed in Part I (Section 10.1).

22.2 How does leaving the affine class remove the obstruction?

Proposition 20.4 settles the second-order question sharply: a quadratic term suffices and an affine repair is impossible. Proposition 20.5 also gives unrestricted formal integration. What remains is the bounded-degree problem: can a given nonzero direction be continued to every order with a fixed bound, such as two, on the v -degree? Which directions lie on an actual polynomial parameter family rather than only a formal one with increasing degrees? These questions cannot be answered merely by taking the formal exponential.

22.3 The local nilpotent algebra at the tame boundary

For the finite coefficient rings Λ_D , determine the nilpotence indices of $g_1, \dots, g_D$ and their mutual products. Corollary 20.6 shows that many are nonzero nilpotents and supplies tangent directions; it does not identify a complete local-ring presentation. A concrete task is to compute a small D presentation, isolate relations forced by the universal obstruction, and prove a pattern by explicit ideal-membership certificates. In particular, does the nilpotence order stay bounded as D increases?

22.4 Higher-degree thickening at a noninjective point

Part I gives a double-point coefficient scheme under an ordinary degree-seven cap with $r = 1 + t + v$ (Theorem 8.1), and asks whether that thickening stabilizes as the degree cap increases (Section 10.3). Our theorem about slope directions at $(v, t, 1)$ does not answer this different question at a noninjective point. Solve the untruncated linearized system there, then its obstruction equations, while keeping the fixed-third-numerator condition explicit.

22.5 Scheme-theoretic gluing of the branches

Field-valued normal forms and tame equivalence do not determine how the noninjective branch approaches the tame locus inside coefficient space. The second-order theorem is a useful constraint: at the simplest tame point, a nonzero first-order slope cannot occur along an honest second-order deformation within the affine class. Determine which higher orders of slope vanishing occur on actual arcs, and how they depend on the remaining first jets. This should be compared with the parameter-plane and boundary calculations in Part I (Theorem 7.1, Section 10.4), without equating a reduced parameterization with the full coefficient scheme.

22.6 Other mixed weights and generalized pole equations

Replace $(-1, 1, 2)$ by $(-1, 1, m)$, or more general primitive mixed-sign triples compatible with a nonsingular linear term. The highest determinant coefficient should give different power relations and equations resembling $L^r L' = \alpha/U^s$. Which exponent pairs force a single critical root, and which permit several? A classification should track polynomiality along the coordinate hyperplane, not just solve the localized equations. The arithmetic of the pole orders may provide a systematic dividing line between rigid and flexible weighted classes. [write] Part I's Section 10.5 asks the analogous question for its Wronskian exponents; here r and s in $L^r L'$ are exponents.

22.7 Positive characteristic under degree hypotheses

Several proof steps fail in characteristic p : derivatives of p th powers vanish, multiplicity drops can change, and coefficients such as $(m+3)(2m+3)/(m+1)$ need not be nonzero. Determine explicit degree ranges and excluded primes for which versions of constant-slope rigidity and the obstruction theorem remain valid. Then construct and classify the first Frobenius-driven failures. Simply reducing the characteristic-zero formulas does not prove an unrestricted positive-characteristic theorem.

22.8 A formal verification route

A compact formalization can be organized around the actual dependencies: weighted monomial encoding and localization; the three determinant coefficients; rational constants of differentiation; valuation factorization; the one-critical-root pole lemma; polynomial divisibility and origin constraints; and reconstruction. The nilpotent obstruction then reduces to coefficient extraction and a leading-term argument. The companion symbolic identities are suitable targets for kernel-checked normalization certificates, but a proof assistant must still verify the all-degree pole and factorization steps. No Lean or Rocq file in this package is presented as already proving them.

22.9 Recognition, sparsity, and arithmetic optimization

The coefficient-only test separates structural classification from coefficient optimization. Over $\mathbb{Q}$, one can now optimize denominator height, integral coefficient height, or collision height within the explicit diagonal-and-shear orbit. Such a project should specify its normalization and cost function: minimizing ordinary degree, number of expanded monomials, coefficient height, and the size of a formal certificate are different problems. Extending the software to sparse arrays and exact algebraic number fields is also useful, but does not change the field theorem.

23 Conclusion of Part II

The passage from a scalar third slope to a polynomial third slope was a real missing case in the repository's weighted classification. It is now excluded without a degree cap. The mechanism is a rational differential invariant whose poles are rigid enough that global polynomiality forces the slope constant. This completes the field-valued classification of all affine-in- z Keller maps with the specified weights and turns it into a direct recognition procedure with concrete inverse or collision certificates.

At the same time, the coefficient schemes retain information that no field-valued normal-form theorem can see. Nonconstant slopes exist over dual numbers in arbitrarily high degrees, and every nonzero slope direction at the simplest tame point has a computable, unavoidable second-order obstruction inside the class. The combination of exact rigidity and explicit nilpotent failure gives both a completed result and a sharply defined next research program.

24 Proof audit and dependency ledger

[write] This section is the manuscript's Appendix A.

Claim	What the proof uses
Constant slope	Characteristic zero; all three numerators affine in v ; polynomial lift; unit normalized origin factors; rational factorization over an algebraic closure; regularity at roots of U ; origin constraints.
Full classification	Constant-slope theorem; the explicit zero-slope calculation; reconstruction and determinant expansion.
Tame/noninjective split	Written inverse and two explicitly distinct ground-field source points; no topological argument.
Degree spectrum	Exact core leading terms and a nonzero source-shear leading term; no enumeration.
Cubic degree and S_3	Direct identities; algebraic independence from nonzero Jacobian; Gauss's lemma; discriminant criterion for a cubic.
Reduced-base rigidity	Field theorem in residue fraction fields; intersection of prime ideals equals the nilradical.
Second-order obstruction	Generic determinant coefficient J_2 ; first-order slope equation; nonzero leading coefficient.
Coefficient-scheme nilpotents	Reduced-base rigidity; dual-number evaluation maps; their linearly independent tangent values.

Potential gaps explicitly avoided. The moving coordinate is used in a rational function field, not declared a global automorphism. The root argument is performed over an algebraic closure and descended only after showing coefficient vanishing. The case $c = 0$ is ruled out before division by c , and the case $g = 0$ is treated separately. A constant Jacobian is not inferred from finite examples. The reduced-base theorem is not extended without qualification to nonreduced rings. The obstruction theorem allows arbitrary second-order coefficients rather than testing only the displayed family. Ordinary polynomial degree is kept separate from geometric degree.

What has not been certified. The written proofs have not been checked by a proof-assistant kernel or an independent referee. The exact computation checks do not establish literature priority, a global minimum degree, a classification beyond the linearity and weight hypotheses, or a presentation of the entire coefficient scheme. The source comparison is restricted to the materials listed in the references and in `SOURCES.md` ([write] shipped as `02-affine-slope-SOURCES.md`).

25 Reproducibility of Part II

[write] The first paragraph is the manuscript's Appendix B, with the delivered paths.

The package contains the complete $\text{\LaTeX}$ source, the compiled PDF, `code/weighted_keller.py`, `code/verify_results.py`, `requirements.txt`, and the recorded check output. From the package root:

```
python -m pip install -r requirements.txt
python code/verify_results.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The article uses standard $\text{\LaTeX}$ packages and embedded Computer Modern family fonts through Latin Modern. No external bibliography processor or network access is required to rebuild the PDF once the packages are installed. The verification scripts perform exact rational and symbolic arithmetic; they do not download or modify the repository.

25.1 Shipped names

[write] In ProveIt the package’s companion files carry the prefix `02-affine-slope-`, the next number in this report’s local sequence (Part I’s files are unprefixd):

Delivered	Shipped
<code>code/verify_results.py</code>	<code>code/02-affine-slope-verify_results.py</code>
<code>code/weighted_keller.py</code>	<code>code/02-affine-slope-weighted_keller.py</code>
<code>code/__init__.py</code>	<code>code/02-affine-slope-__init__.py</code>
<code>data/verification.json, .txt</code>	<code>data/02-affine-slope-verification.json, .txt</code>
<code>requirements.txt</code>	<code>data/02-affine-slope-requirements.txt</code>
<code>SOURCES.md, STATUS.md</code>	<code>02-affine-slope-SOURCES.md, 02-affine-slope-STATUS.md</code>

They are byte-identical to the delivery. The manuscript, its README and its PDF are not shipped: this Part is the manuscript, and the report’s `article.pdf` is a build of this text.

25.2 Rerunning the checks

[write] Do not run the prefixed script in place. It imports its helper by the delivered name (`from weighted_keller import`), which does not exist under the shipped names, so it stops with an `ImportError`; and if that import were satisfied it would write `data/verification.json` and `data/verification.txt` next to the `code/` directory, which in this report are *Part I*’s records. The package’s `__init__.py` also makes `code/` a package that shadows Python’s standard module `code`. Rerun it on a copy with the delivered names instead:

```
mkdir -p <scratch>/pkg/code
cp code/02-affine-slope-verify_results.py <scratch>/pkg/code/verify_results.py
cp code/02-affine-slope-weighted_keller.py <scratch>/pkg/code/weighted_keller.py
cp code/02-affine-slope-__init__.py <scratch>/pkg/code/__init__.py
cp data/02-affine-slope-requirements.txt <scratch>/pkg/requirements.txt
cd <scratch>/pkg && python code/verify_results.py
```

It writes `data/verification.json` and `data/verification.txt` inside the copy; compare them with the shipped `data/02-affine-slope-verification.json` and `.txt`. At the write this was done from the shipped files with SymPy 1.14.0 (under `uv`, Python 3.13.5) in about 11 seconds: all twelve groups passed and both records equal the shipped ones apart from line endings. The JSON embeds the Python version, so a run under another version differs in that field.

A An explicit presentation of the coefficient ideal

For readers who prefer generators to the vector formula (8.5), Theorem 8.1 says that I is exactly the ideal generated by

$$\begin{aligned}
 & q_{40}, \quad (q_{21} - 4)^2, \\
 & 144p_{20} + 7q_{21} + 100, \\
 & 8p_{11} + q_{21} + 12, \\
 & 216p_{30} - 27q_{21} - 116, \\
 & 6p_{21} - q_{21} - 4, \\
 & 54p_{40} + 3q_{21} + 4, \\
 & 54p_{31} + 3q_{21} + 4, \\
 & 8q_{01} - 9q_{21} - 36, \\
 & 8q_{20} + 11q_{21} + 20, \\
 & 4q_{11} + 9q_{21} + 12, \\
 & q_{30} - q_{21}.
 \end{aligned} \tag{A.1}$$

These equations apply over every $\mathbb{Q}$ -algebra, not just over fields. Over a field the square forces $q_{21} = 4$, and all remaining coefficients are the point (8.6). Over the dual numbers, $q_{21} = 4 + \varepsilon$ produces exactly the tangent vector (8.7).

As a useful independent check of the equation reconstruction, the coefficients corresponding to the last three monomials t, v^2, v in (8.8) are, respectively,

$$\begin{aligned}
 f_{16} &= -p_{11} + 2p_{20}q_{01} - 2q_{20} - 2, \\
 f_{17} &= 2p_{11}q_{01} - 3q_{11}, \\
 f_{18} &= p_{11}q_{01} + q_{01} - q_{11} - 3.
 \end{aligned}$$

For the first two monomials they are

$$f_1 = -5p_{31}q_{40} - 4p_{40}q_{40}, \quad f_2 = -9p_{31}q_{40}.$$

The remaining equations are recovered by the same determinant expansion; the complete list is included in the JSON certificate.

B Small exact certificates for the deformation calculation

The primitive integral tangent vector is $n = 144u$. To keep the second-order residual integral, define

$$\hat{H}_i = [e^2]f_i(c_* + en) = 144^2 H_i.$$

In the ordering (8.8), the nonzero data and the cokernel functional are as follows. Zeros are included to make the ordering unambiguous.

Row	Monomial	$\widehat{H}_i$	L_i
1	t^7	0	-205
2	t^6v	0	0
3	t^6	0	-46
4	t^5v	0	0
5	t^5	1008	-40
6	t^4v	1872	24
7	t^4	-4140	-8
8	t^3v^2	864	0
9	t^3v	-9504	4
10	t^3	2772	0
11	t^2v^2	-5184	0
12	t^2v	12636	0
13	t^2	1620	0
14	tv^2	9720	0
15	tv	-2592	0
16	t	-2268	0
17	v^2	-5832	0
18	v	-2916	0

The obstruction is visible in four integer products:

$$(-40)(1008) + 24(1872) + (-8)(-4140) + 4(-9504) = -288.$$

Dividing by 144^2 gives $LH = -1/72$.

For the nonzero minor in (8.11), the zero-based row indices in the JSON matrix are

$$(0, 2, 4, 5, 6, 9, 11, 12, 13, 15, 17),$$

and the zero-based column indices are

$$(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11).$$

Thus an independent checker only needs the reconstructed derivative matrix, one 11×11 determinant, and the two matrix-vector products Mu and LM . The companion JSON stores the matrix itself as rational strings for convenient cross-checking.

C Reproduction and source ledger

The supplied archive contains the following files:

File	Purpose
<code>article.tex</code>	Self-contained LaTeX source, including bibliography.
<code>article.pdf</code>	Compiled article.
<code>verify_results.py</code>	Reconstructs and checks all finite identities.
<code>verification.json</code>	Exact equations, matrix, and certificate data.
<code>verification.txt</code>	Output of the successful verification run.
<code>README.md</code>	Build commands, scope, and trust boundary.
<code>requirements.txt</code>	Reproducible SymPy dependency.

Build the article with two runs of `pdflatex article.tex`, or with `latexmk -pdf article.tex`. Run the checks with

```
python verify_results.py --output verification.json
```

[write] In ProveIt the script is `code/verify_results.py`; the records and `requirements.txt` are in `data/`; the delivered PDF is not shipped (`article.pdf` is a build of this text); the delivered README is replaced by the report README. The command above is the delivery’s; its default output `verification.json` is relative to the working directory, so in ProveIt pass an output path outside the report (Appendix D.3).

The verification does not download or execute any repository code. The repository is a cited mathematical source; its original map and the relevant formulas are reconstructed in the script. The source ledger is:

Source category	Role in the report
Repository map	Original map, its known collision, and the finite-search baseline; not new claims of this article.
External research	Context for graded maps and examples outside our class; not dependencies of the classification proof.
Standard algebra	Nullstellensatz and Nakayama’s lemma in the passage from an isolated obstructed tangent to a double point.
This article	All-degree classification, family closure, reduced rigidity, and the full degree-seven coefficient-algebra identification.

D Provenance of this report

D.1 One manuscript

[write] This report is built from *one* manuscript: manuscript 01 of batch 36 of ProveIt’s incoming reports, the archive `ProveIt_Weighted_Keller_Rigidity` (inner directory of the same name, main file `article.tex`, a 23-page PDF dated 24 September 2026, PDF metadata author “ChatGPT; prepared for Vladimir Reshetnikov”), pinned to ProveIt commit `e21766d04c2b8a9b2cd43e563024b3bd1b9`. The archive arrived in commit `e13affd32` and was placed in `1a1396d4d` as a new report. Nothing was merged and nothing was selected away: every result, proof, example, table, question and limitation of the manuscript is printed. The manuscript contributed everything except the text marked [write]. [Added 29 September 2026, batch 44: this appendix describes Part I. The report is now built from *two* manuscripts; Part II is batch-44 manuscript 04, pinned to `9b24a3a8d545af9624f6ac455f5b548be62818b6`, and its provenance, shipped files and editorial changes are in Sections 12.1, 12.4 and 25.]

The editorial changes are these.

- (1) Every label carries the prefix `wkr:`. The source’s 96 labels are kept, unchanged after the prefix. Six were added: `wkr:sub:project`, `wkr:sub:gao`, `wkr:app:report`, `wkr:app:onesource`, `wkr:app:pinned` and `wkr:app:rerun`.
- (2) Sections 1.4 and 1.5 are new and end Section 1, so no source number moved. They credit the project’s weighted determinant identity and collision mechanism, state the project’s formal status, and correct the reading of Gao’s “degree four”.
- (3) Notes marked [write] were added on the title page, after the proof of Proposition 2.2, after the collision family (6.7), in Sections 9 and 10, and in Appendix C; this appendix is

new. The PDF bookmark of the title page was fixed (no duplicate page anchor), and the command-line option `--output` in Appendix C is now typeset with two hyphens (the delivered PDF showed one dash).

No mathematical statement of the manuscript was changed and no symbol was renamed; the added notes write α for the project’s coefficient $a = [t^2]p$, because a is a family parameter here. The report’s $t = xy$ agrees with the project’s research notes but not with the sibling report on arithmetic fibers, where $t = y + 1/x$.

D.2 Repository statements at the pin

[write] The source read the project README, its research notes and `verify_equivariant_sparsity.py` at its pin. The project did not change between the pin and the placement commit. Its statements about the project are accurate: the map, determinant and collision are the project’s (and formalized, Section 1.4); the sparsity script proves that eleven coefficients cannot vanish and identifies the branch $q_{40} = 0$ (script lines 110–127 and 164–167); its certificate contains $(q_{21} - 4)^2$ with the comment that the square “records a harmless nonreduced structure” (lines 129–132), and neither README identifies the scheme as a double point. The one omission, the credit for the determinant identity and the collision mechanism, is repaired in Section 1.4. Shaska [5] and the Stacks Project references were not checked at the write; Gao [6] was checked for Theorems 3.3 and 3.5 (Section 1.5).

At the write, the ideal statement of Theorem 8.1 was also checked in both directions, which the shipped script does not do: with SymPy 1.14.0 the eighteen coefficient equations and the twelve generators of (A.1) have the same reduced Gröbner basis (grevlex), and each set lies in the ideal of the other. The normalization (3.4) was rechecked the same way.

D.3 Rerunning the checks

[write] The script writes its JSON record to `--output`, default `verification.json` in the current directory. Run it from the report with an output path outside the repository, so that neither the report root nor `data/` is written:

```
python code/verify_results.py --output <scratch>/verification.json
```

It needs Python 3.10 or later and SymPy (data/requirements.txt pins `sympy==1.14.0`). At the write it was run so with Python 3.13.5 and SymPy 1.14.0: all thirteen check groups passed, the JSON record equals `data/verification.json` apart from line endings, and the printed output equals `data/verification.txt` apart from line endings and the certificate path in its last line. The record embeds the Python version, so a run under another version differs in that field.

References

- [1] Vladimir Reshetnikov, *ProveIt: Algebra/JacobianConjecture/README.md*. Repository snapshot `e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9`, inspected September 24, 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/Algebra/JacobianConjecture/README.md>.
- [2] Vladimir Reshetnikov, *ProveIt: Algebra/JacobianConjecture/Research/README.md*. Same repository snapshot and inspection date. <https://github.com/VladimirReshetnikov/ProveIt/blob/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/Algebra/JacobianConjecture/Research/README.md>.

- [3] Vladimir Reshetnikov, *verify_equivariant_sparsity.py*. Same repository snapshot and inspection date. https://github.com/VladimirReshetnikov/ProveIt/blob/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/Algebra/JacobianConjecture/Research/verify_equivariant_sparsity.py.
- [4] Vladimir Reshetnikov and contributors, *ProveIt: Algebra/JacobianConjecture*, repository development and research notes, inspected at commit 9b24a3a8d545af9624f6ac455f5b548be62818b6, 29 September 2026 (cited by Part II). <https://github.com/VladimirReshetnikov/ProveIt/tree/9b24a3a8d545af9624f6ac455f5b548be62818b6/Algebra/JacobianConjecture>. The formal status asserted by the repository concerns its own declarations, not the new theorems of Part II.
- [5] T. Shaska, *Graded Keller maps and the Jacobian Conjecture*, arXiv:2607.20210, 2026. Version 2 dated July 25, 2026; consulted September 24, 2026. <https://arxiv.org/abs/2607.20210>. Part II consulted the HTML of the same version on September 29, 2026, for graded coordinates, the known counterexample and coefficient-scheme geometry.
- [6] Shuhong Gao, *Counterexamples to the Jacobian conjecture in dimensions greater than two*, arXiv:2608.00222, 2026. Version 1 dated July 31, 2026; consulted September 24, 2026. <https://arxiv.org/abs/2608.00222>. Part II consulted the HTML of the same version on September 29, 2026, for broader constructions and generic fiber degrees; it is not used there as a classification theorem.
- [7] The Stacks Project Authors, *The Stacks Project*, Theorem 10.34.1, Hilbert Nullstellensatz, Tag 00FV. Consulted September 24, 2026. <https://stacks.math.columbia.edu/tag/00FV>.
- [8] The Stacks Project Authors, *The Stacks Project*, Lemma 10.20.1, Nakayama's lemma, Tag 00DV. Consulted September 24, 2026. <https://stacks.math.columbia.edu/tag/00DV>.